

MAHIDOL UNIVERSITY

Faculty of Information and Communication Technology



Lecture 5
Analog Transmission

September 2025

ปัญญาของแผ่นดิน Wisdom of the Land

มหาวิทยาลัยมหิดล
Mahidol University

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Quote of the Week

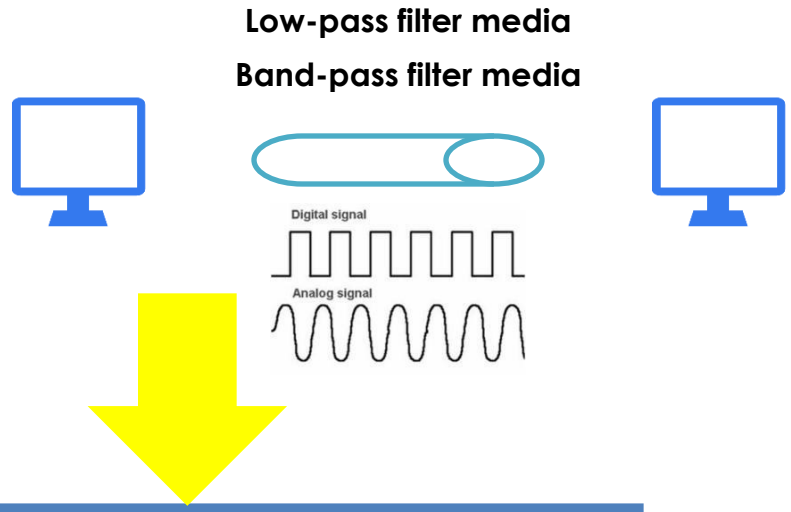
"It always seems impossible until it's done."

— Nelson Mandela



Recap: Data to Signal Conversion

- There are two types of media
  - **Low-pass filter media** (e.g., LAN cable, Fiber optics cable)
    - Allow signal of many frequencies to pass.
    - Suitable for digital transmission
  - **Band-pass filter media** (e.g., telephone line, air)
    - Allow signals with specific frequencies to pass.
    - Suitable for analog transmission.

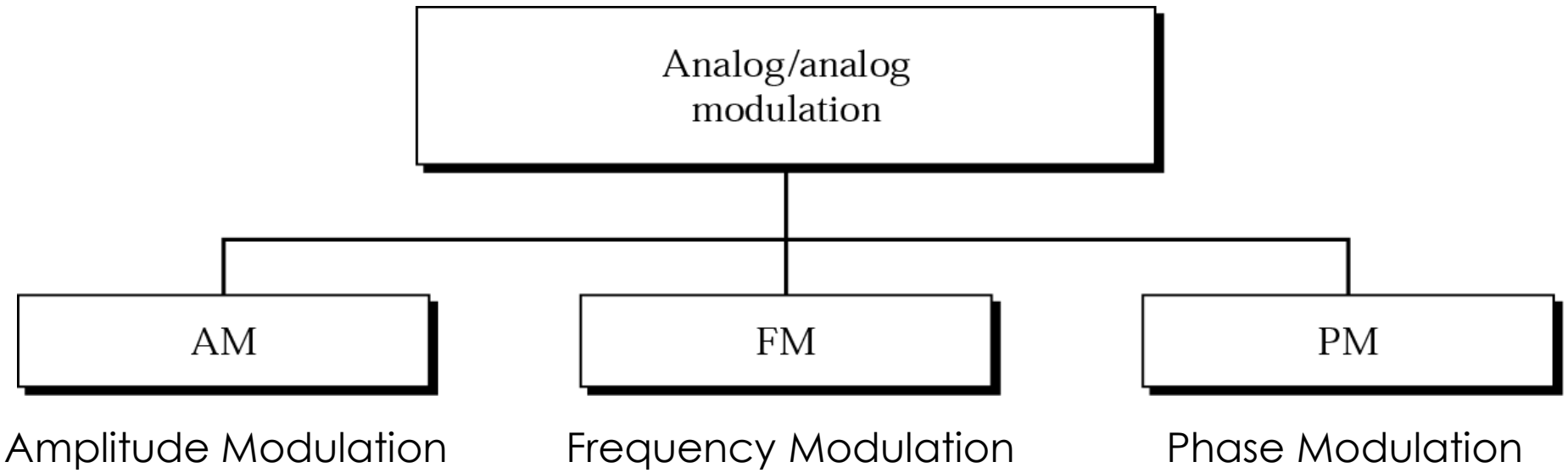


Output / Input	Digital Signal	Analog Signal
	Digital Transmission Low-pass filter media (support digital signal)	**Analog Transmission** Band-pass filter media (support analog signal)
Digital data	**Digital-to-Digital conversion** (Line coding)	**Digital-to-Analog conversion** (Shift keying)
Analog data	**Analog-to-Digital conversion** (Pulse Code Modulation)	**Analog-to-Analog conversion** (Modulation)

Analog-to-Analog Conversion

- Digital transmission **is not** possible in band-pass channels (such as air and telephone cable). Digital transmission **requires infinite bandwidth**.

- Analog signals have to be converted or transformed into the frequency bands that the band-pass channels allow to go through.



Modulation

- Modulation is a technique that changes one of the characteristics of the carrier.
  - Amplitude modulation: change the amplitude of the carrier frequency
  - Frequency modulation: change the frequency of the carrier frequency
  - Phase modulation: change the phase of the carrier frequency

$$y = \boldsymbol{A}\sin(2\pi \boldsymbol{f} t + \emptyset) \;\; = A\sin(\omega t + \varphi)$$

2¶f = ω →f =ω/2¶ Hz

Exercise

- A signal can be represented as $y = 20\sin(500\,t + ¶/2)$

- What is the frequency of this signal in Hertz?

$$y = \boldsymbol{A}\sin(2\pi \boldsymbol{f} t + \emptyset)$$

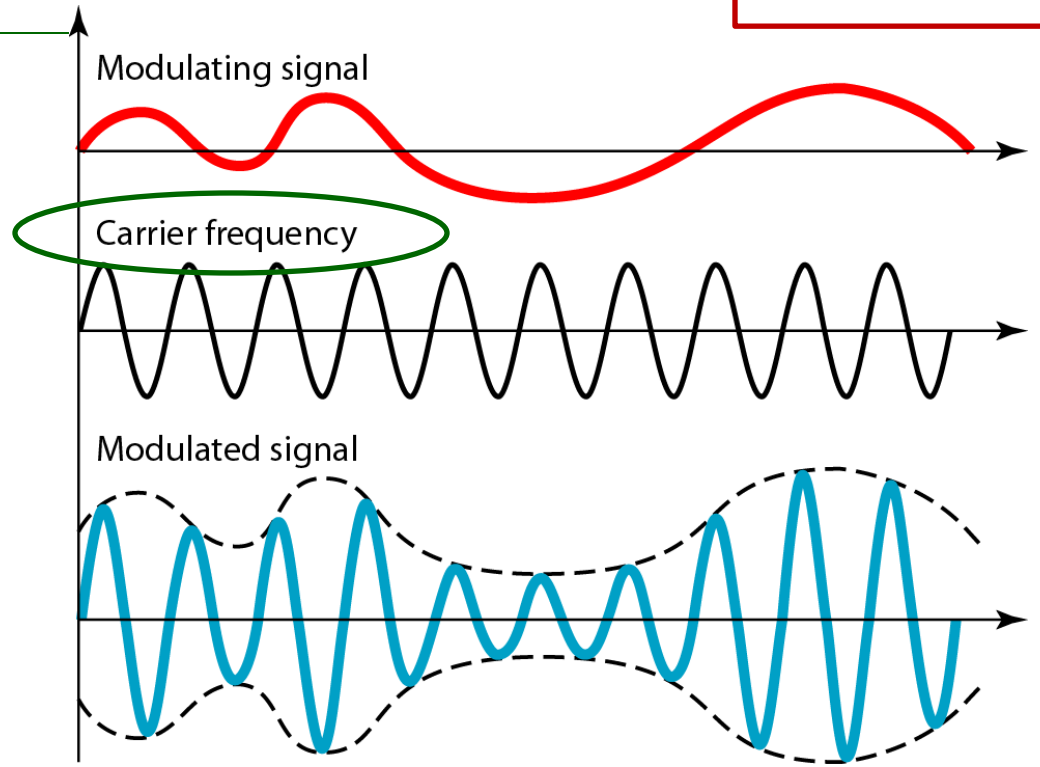
$2\pi f = 500 \longrightarrow f = 250/\pi$

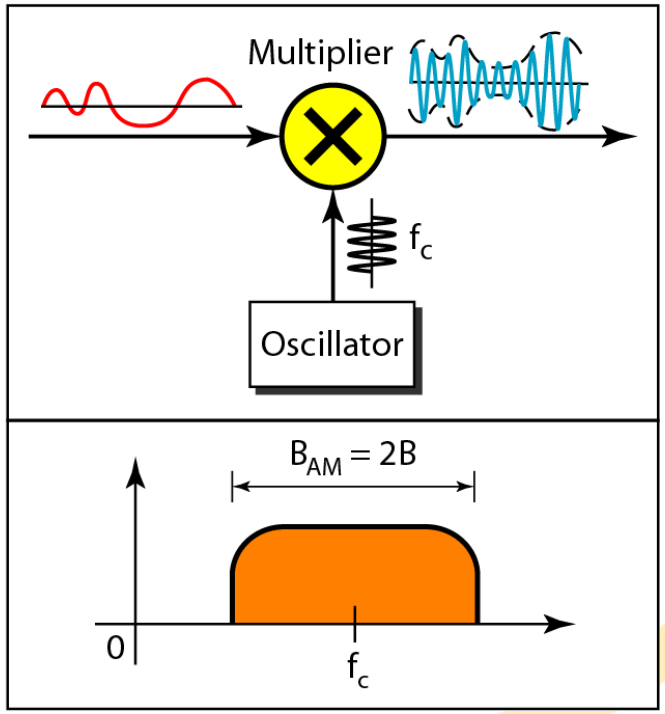


Amplitude Modulation (AM)

- The carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal
  - The carrier signal acts like a sampling procedure
  - Frequency and phase remain the same
- The total bandwidth required is $B_{AM} = 2 \times B_m$

Modulating Signal





Exercise

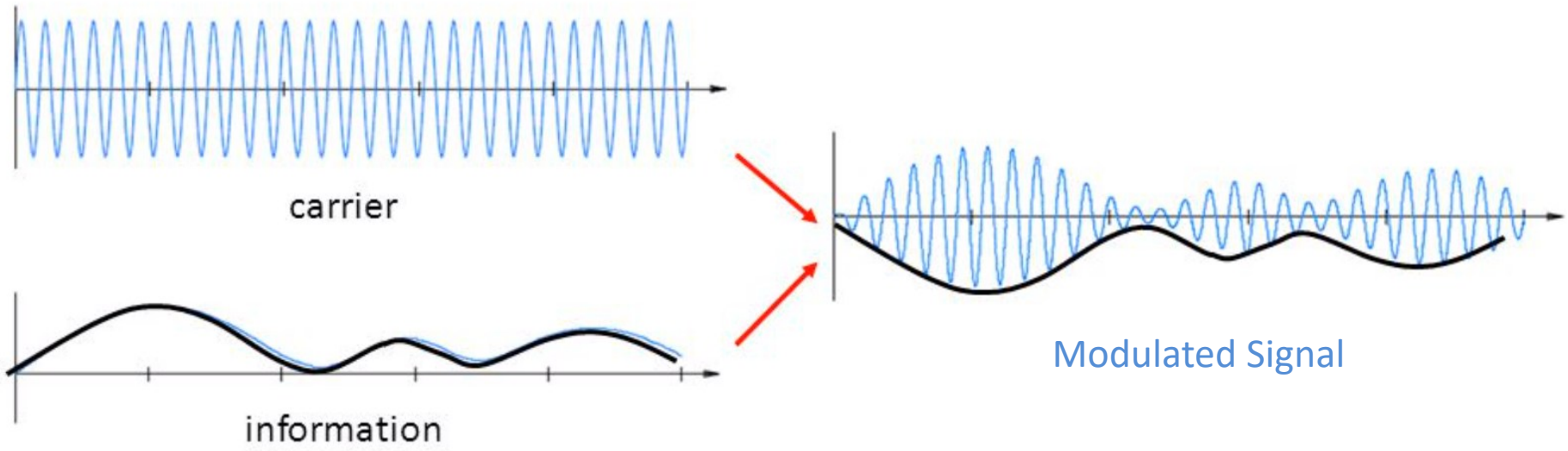
- If the carrier frequency (fc) = 200 Hz
- And the modulating signal has the frequency = 50 Hz
- The total bandwidth required = _____ Hz
- What should be the lower frequency and the upper frequency of the AM modulated signal?
  - Lower frequency = _____ Hz
  - Upper frequency = _____ Hz

Exercise2

- If the carrier frequency (fc) = 200 Hz
- And the modulating signal has the frequency = 40 Hz
- The total bandwidth required = [ ]
- What should be the lower frequency and the upper frequency of the AM modulated signal?
  - Lower frequency = [ ] Hz
  - Upper frequency = [ ] Hz

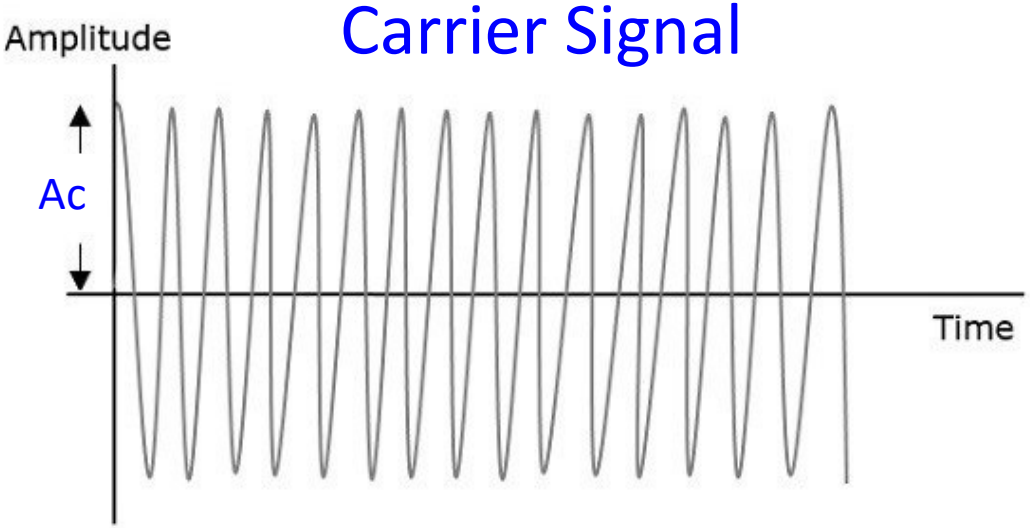
Amplitude Modulation (AM)

- The amplitude of a carrier is modified in proportion to the information signal.
- The frequency of the carrier is fixed.

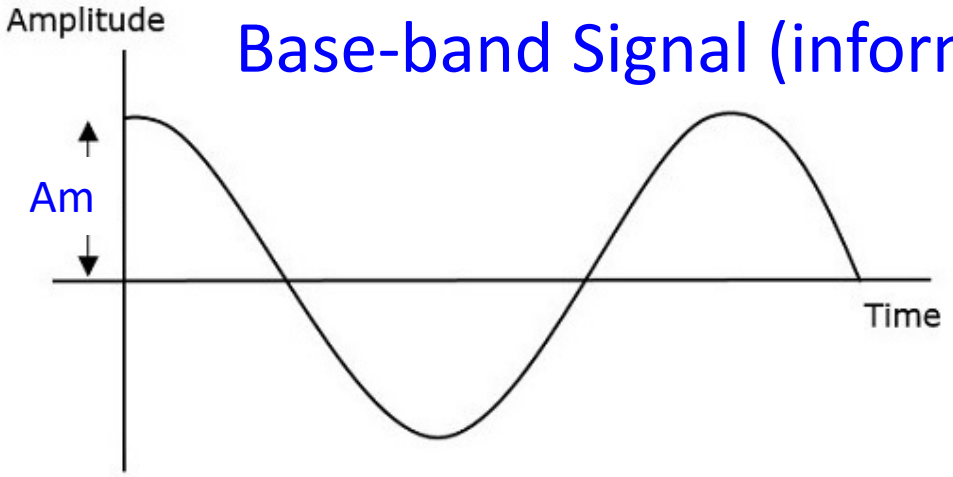


New Amplitude = Ac + Am

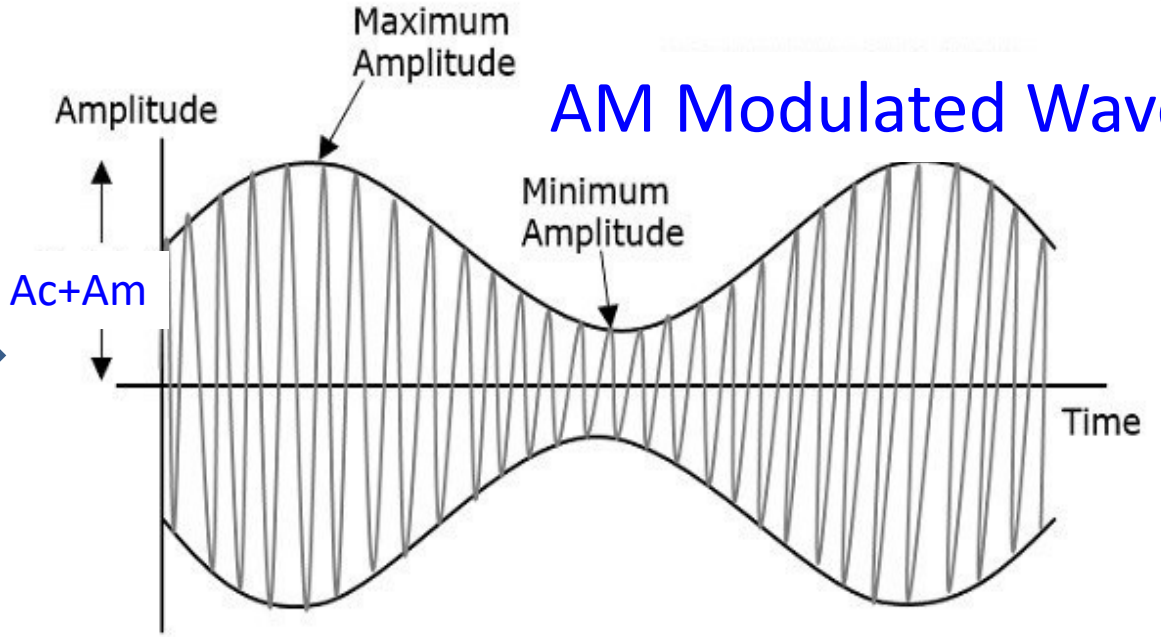
Carrier Signal



Base-band Signal (information)



AM Modulated Wave

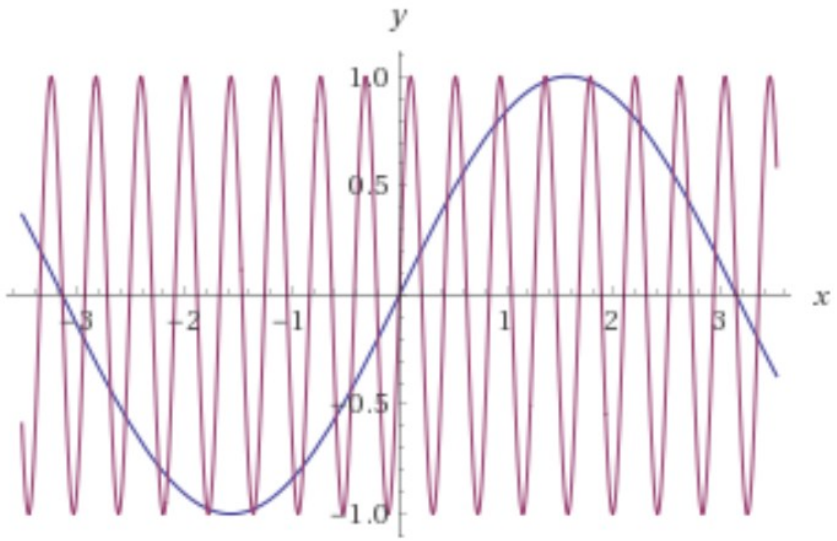


For the modulation ratio or modulation index = 100%

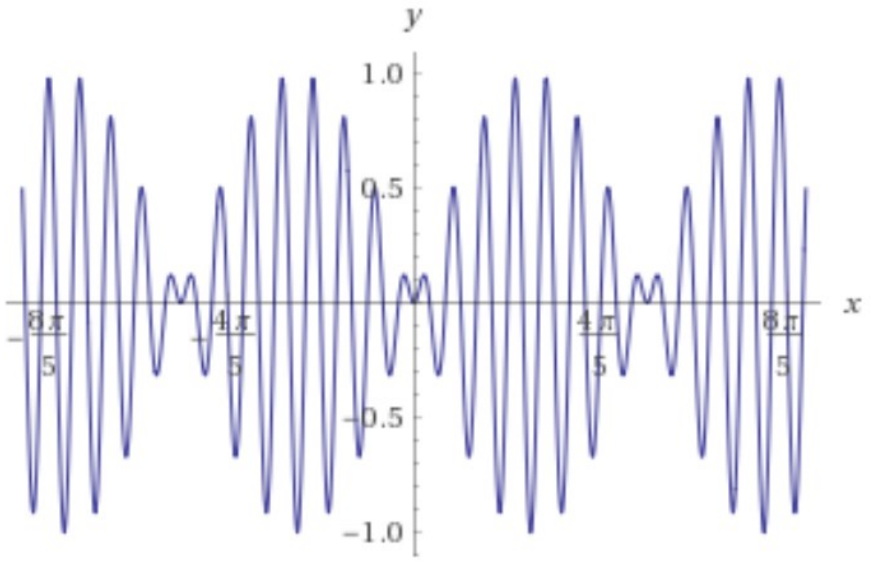
https://www.tutorialspoint.com/analog_communication/analog_communication_amplitude_modulation.htm

Amplitude Modulation

- Example: The modulation is the multiplication of the carrier signal and the information.
  - Carrier: sin(15x)
  - Information: sin(x)

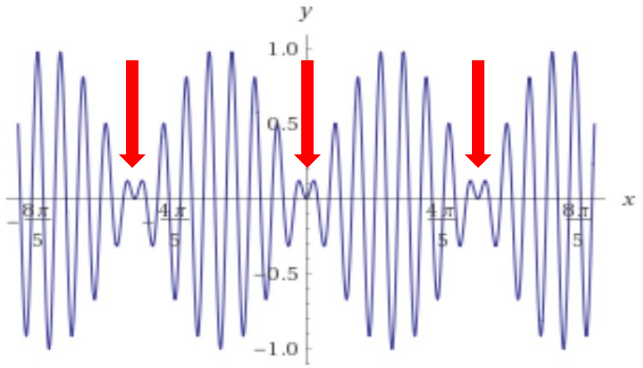


Modulated signal: sin(15x)*sin(x)



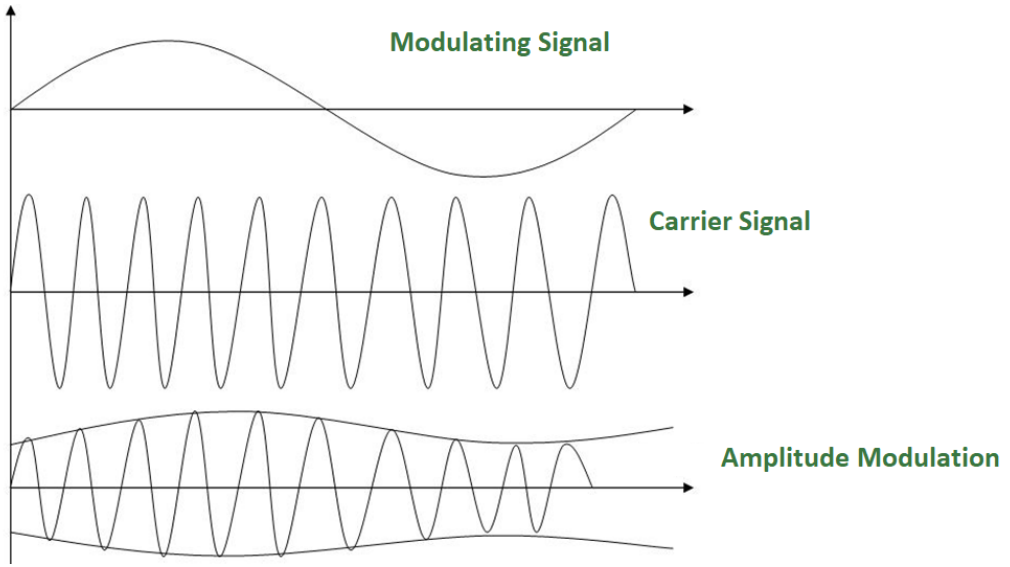
Problems of Amplitude Modulation

- Power level to zero

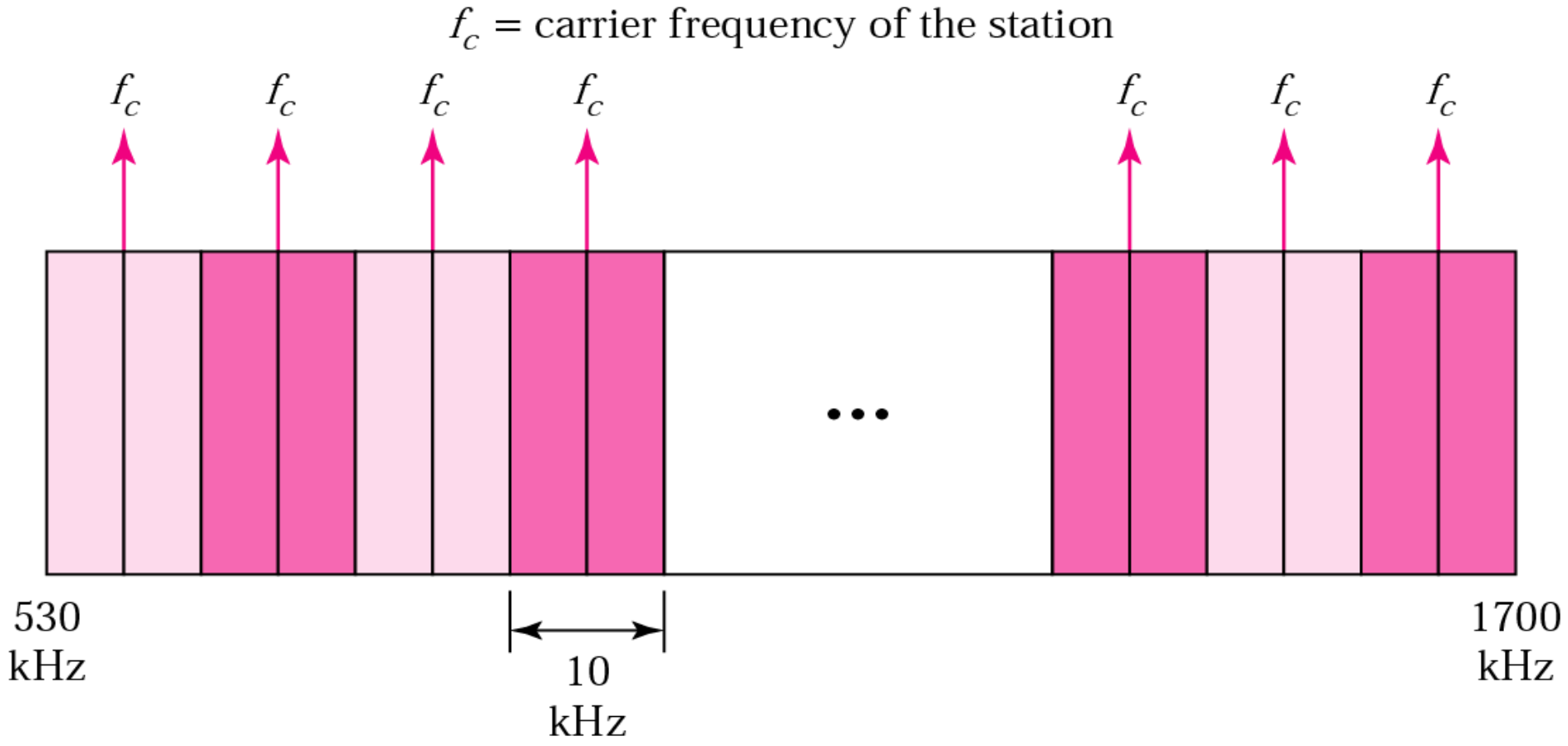


- Practical systems do not allow for a modulated signal to approach zero. Thus, in practice, modulation only changes the amplitude of a carrier slightly in order to maintain high the carrier energy.

- Example:



AM Radio Bandwidth Allocation



Example

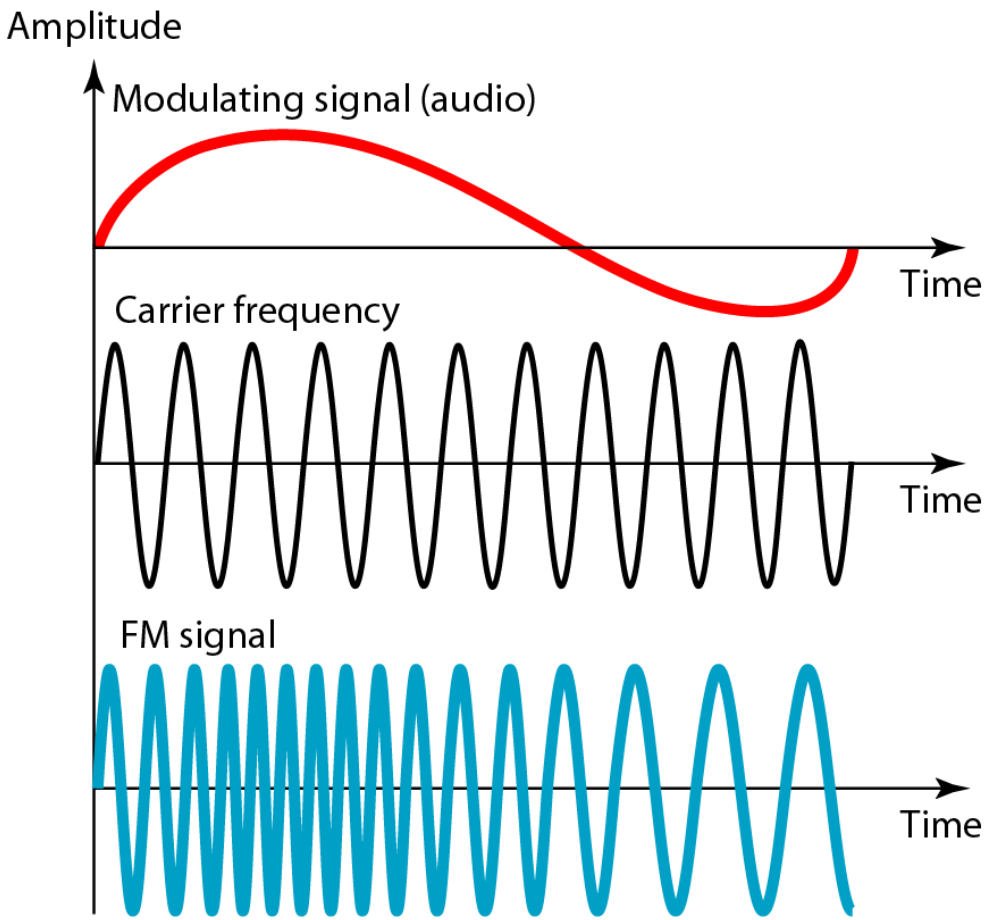
- We have an audio signal with a bandwidth of 4 KHz. What is the bandwidth needed if we modulate the signal using AM? Ignore FCC regulations.

- Answer: Bandwidth required is 2 times the bandwidth of the original modulating signal

  - BW =

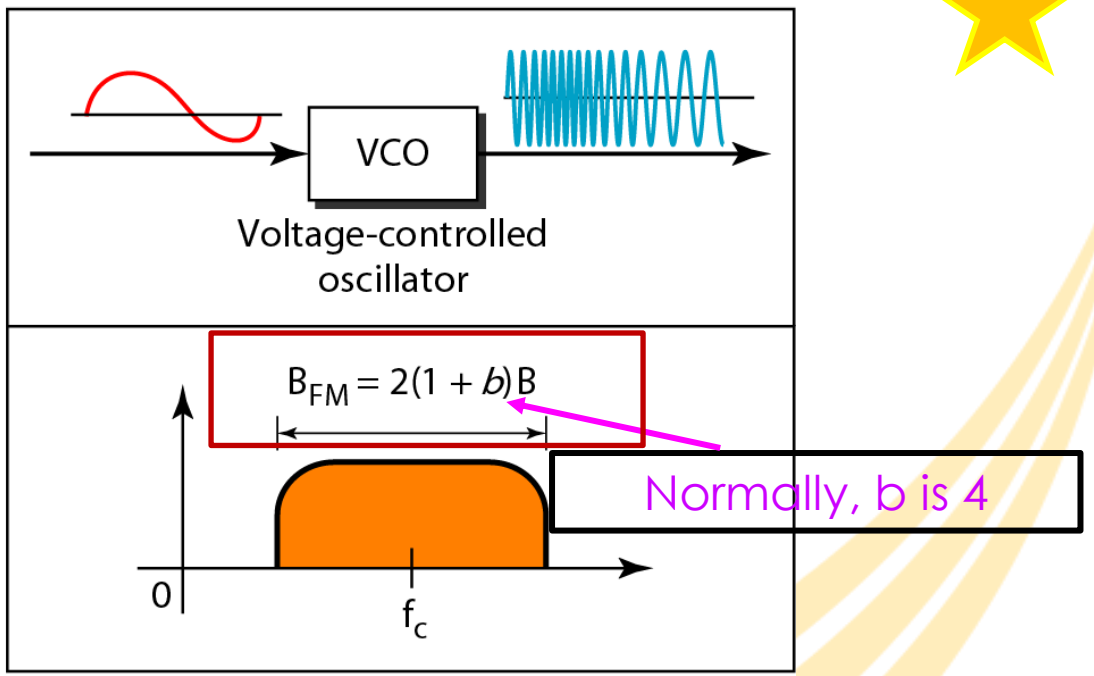
Frequency Modulation (FM)

- Frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal
  - Peak amplitude and phase of the carrier signal remain constant



Bandwidth required == 10 times !

Bfm = 10 B



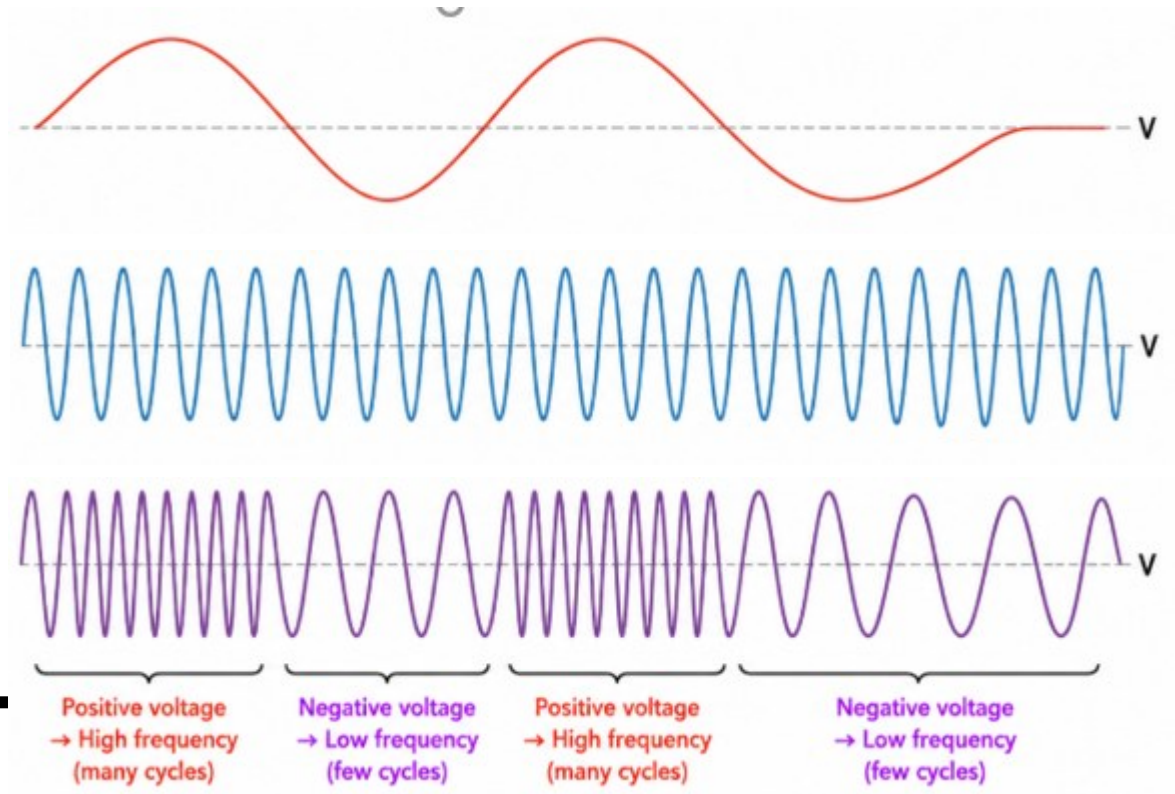
Exercise

- Draw an FM signal resulting from the modulation of the following analog data and carrier signals.

Analog data
(modulating signal)

carrier signal

FM signal
(frequency varies)



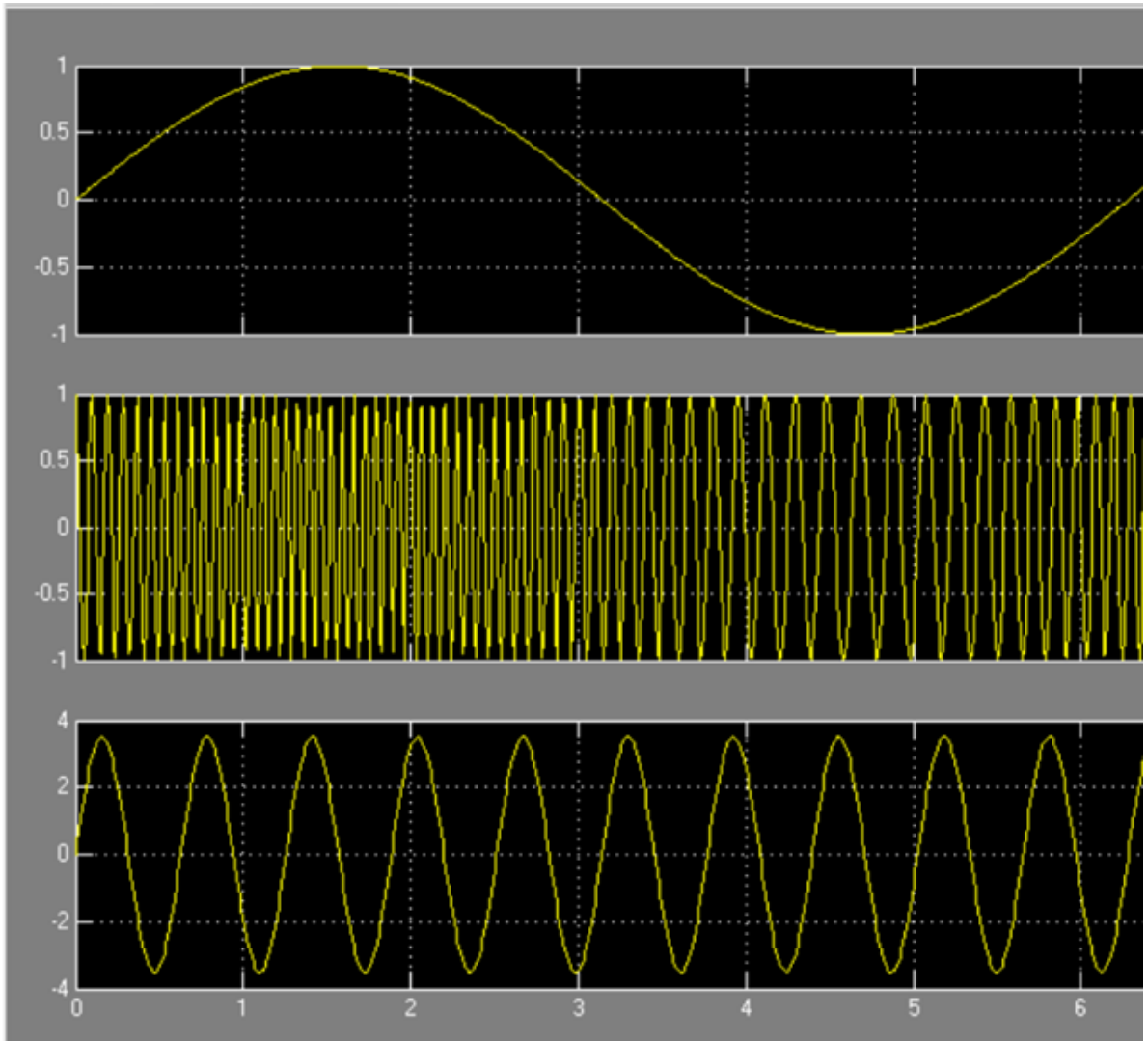
Assume the positive voltage will have a high frequency (many cycles), and the negative voltage will have a low frequency.
(similar to the figures posted below)

Example for Ex 4

Message

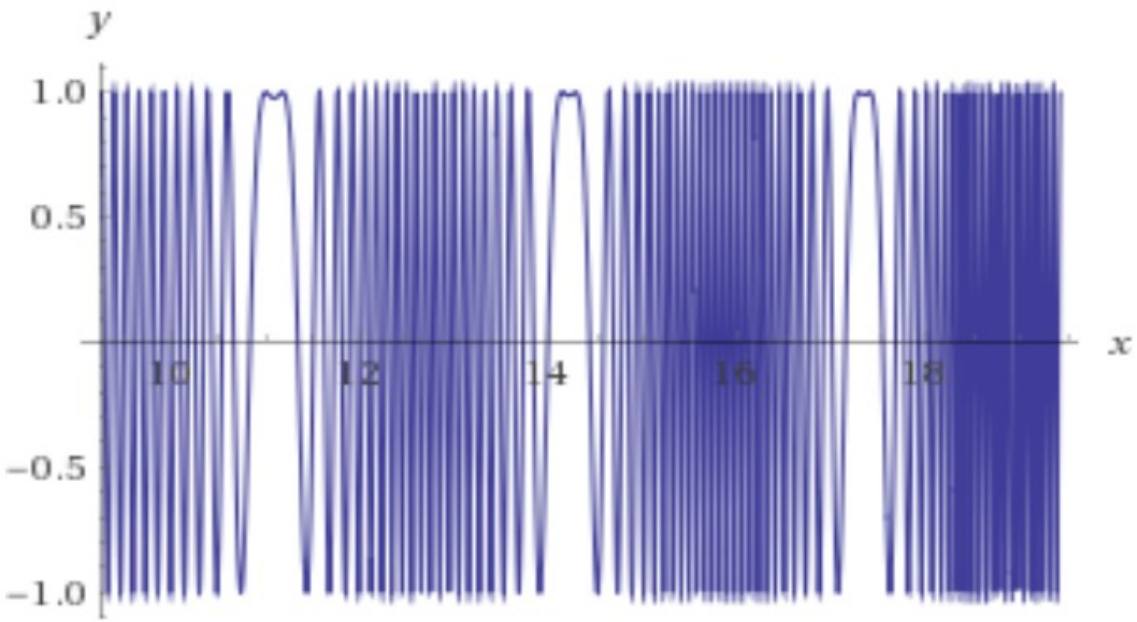
Frequency Modulated Signal

Carrier

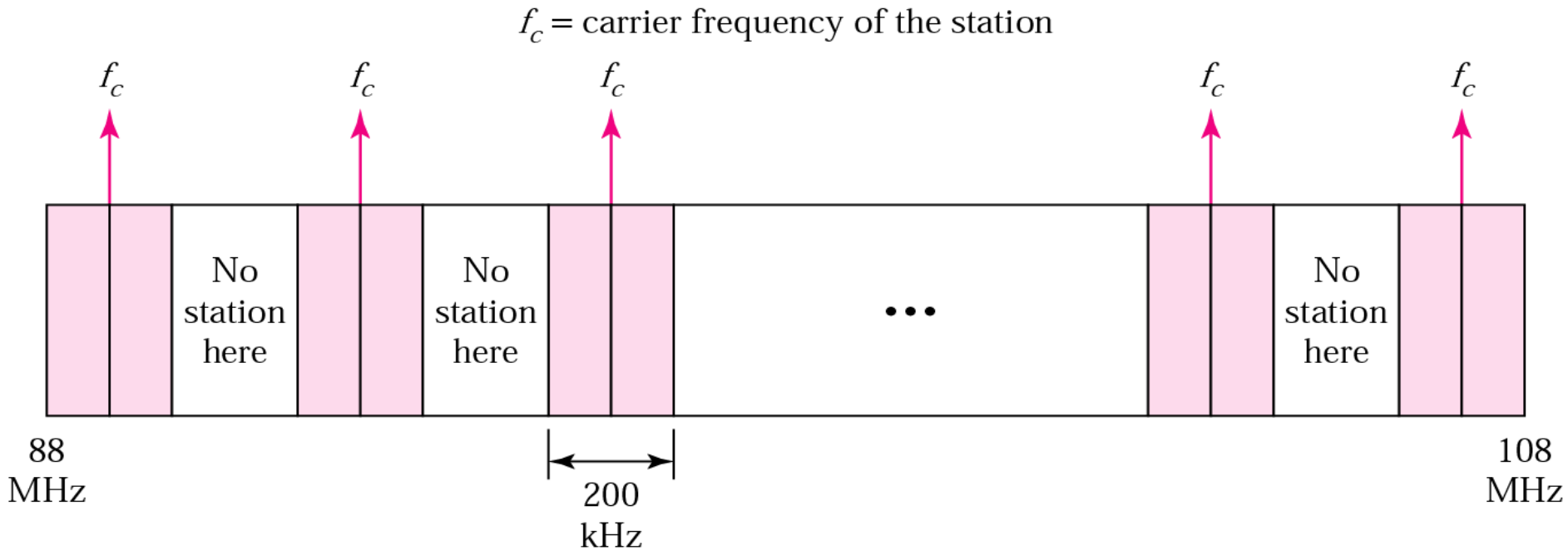


Frequency Modulation

- Signal amplitude can be easily affected by the environment.
- Signal frequency however, is quite **stable**.
- In FM, frequency of the carrier changes according to the information.
- Example: sin(sin(x) * 5x)



FM Band Allocation



Example

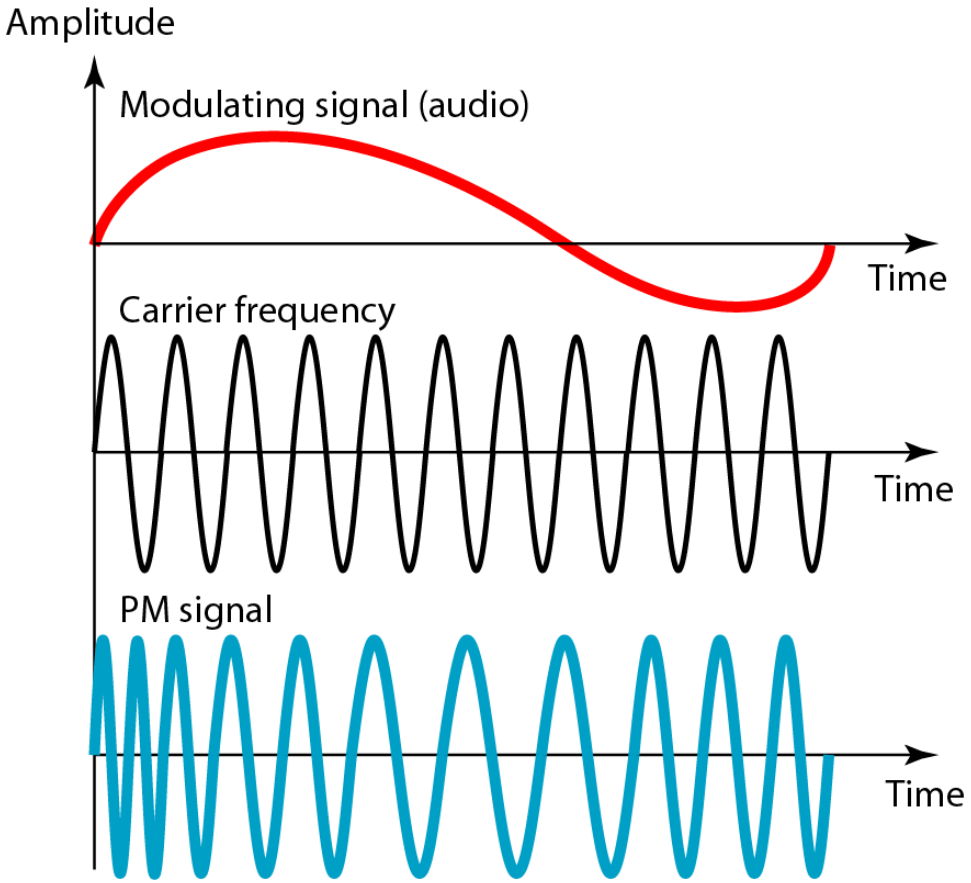
- We have an audio signal with a bandwidth of 4 MHz. What is the bandwidth needed if we modulate the signal using FM? Ignore FCC regulations.

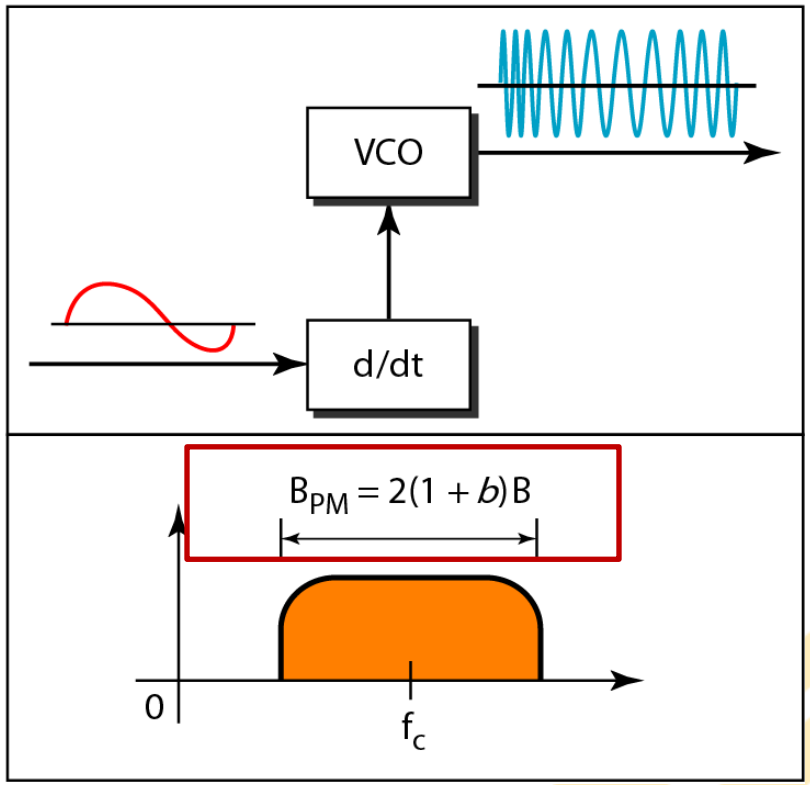
- Answer: An FM signal requires 10 times the bandwidth of the original signal. We have

BW =

Phase Modulation (PM)

- Due to simpler hardware requirements, PM is used in some systems as an alternative to FM.
- The phase of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal



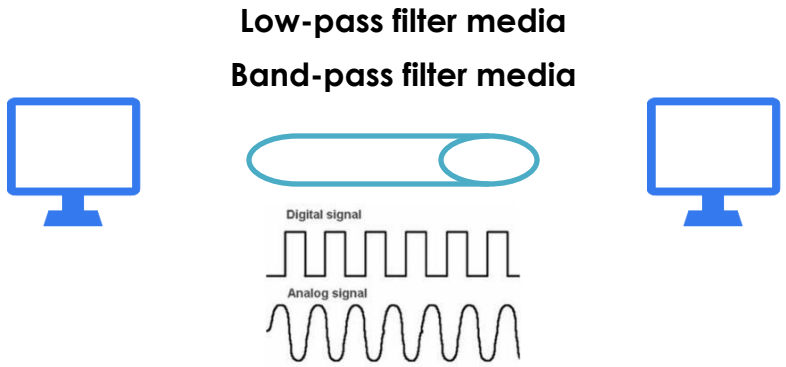


d/dt is the differential *operator*

FM & PM have the same formula

Data to Signal Conversion

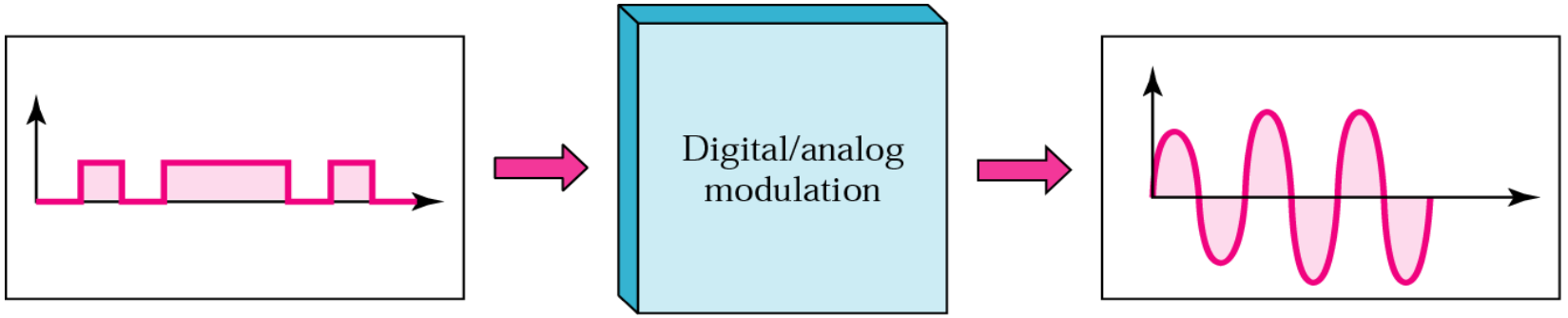
- There are two types of media
  - **Low-pass filter media** (e.g., LAN cable, Fiber optics cable)
    - Allow signal of many frequencies to pass.
    - Suitable for digital transmission
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    - Allow signals with specific frequencies to pass.
    - Suitable for analog transmission.

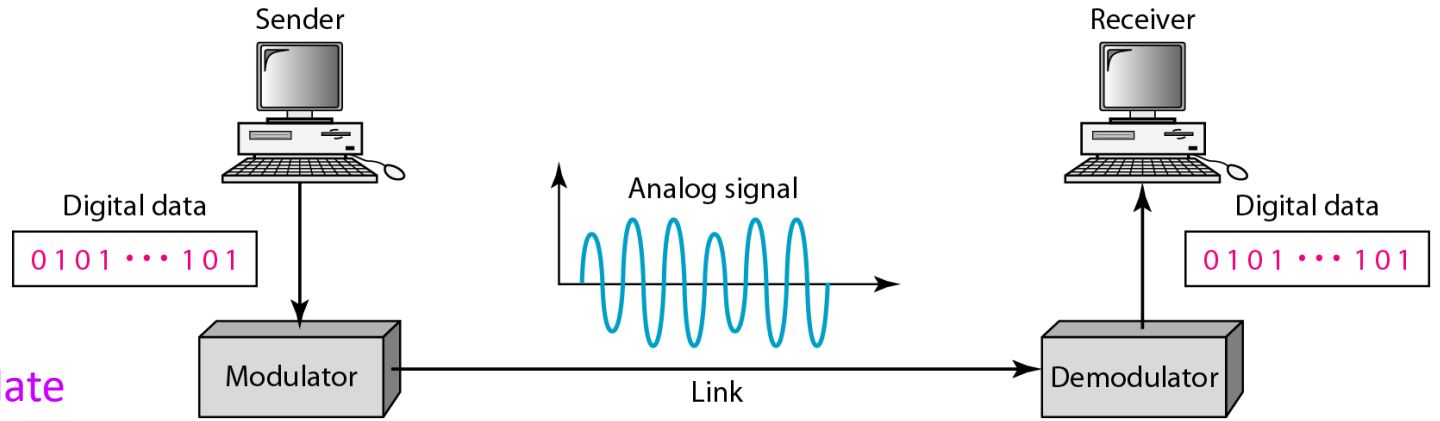


Input \ Output	Digital Signal	Analog Signal
	Digital Transmission Low-pass filter media (support digital signal)	**Analog Transmission** Band-pass filter media (support analog signal)
Digital data	**Digital-to-Digital conversion** (Line coding)	**Digital-to-Analog conversion** (Shift keying)
Analog data	**Analog-to-Digital conversion** (Pulse Code Modulation)	**Analog-to-Analog conversion** (Modulation)

Digital-to-Analog Conversion

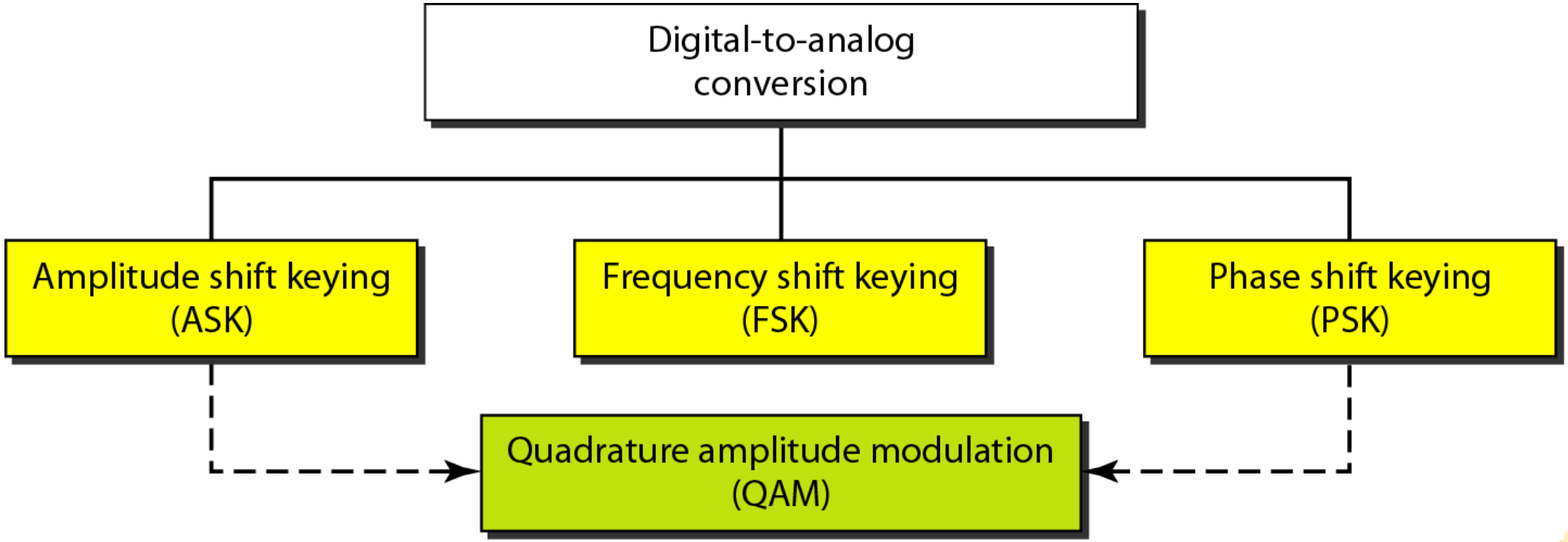
- In case that the channel available can only support analog transmission but the data to be sent are in the form of digital data, we have to transmit analog signal providing digital data.
- Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.





Modem =
Modulate + Demodulate

Digital-to-Analog Conversion Technique



Shift Keying

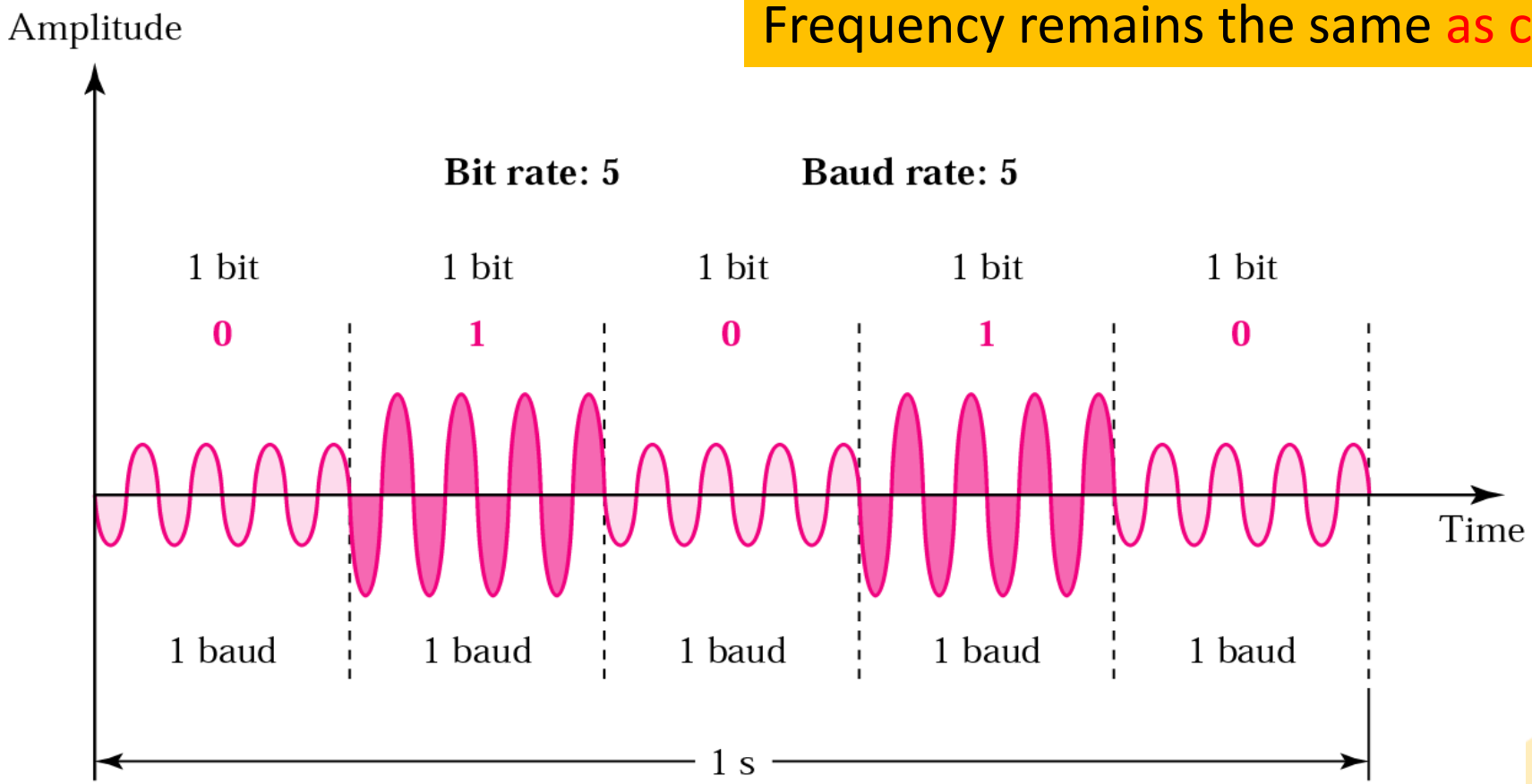
- Instead of a continuum of possible values, digital shift keying has a fixed set of possible values to be shifted to.
  - Mapping to the power levels of a digital signal.



Amplitude Shift Keying (ASK)

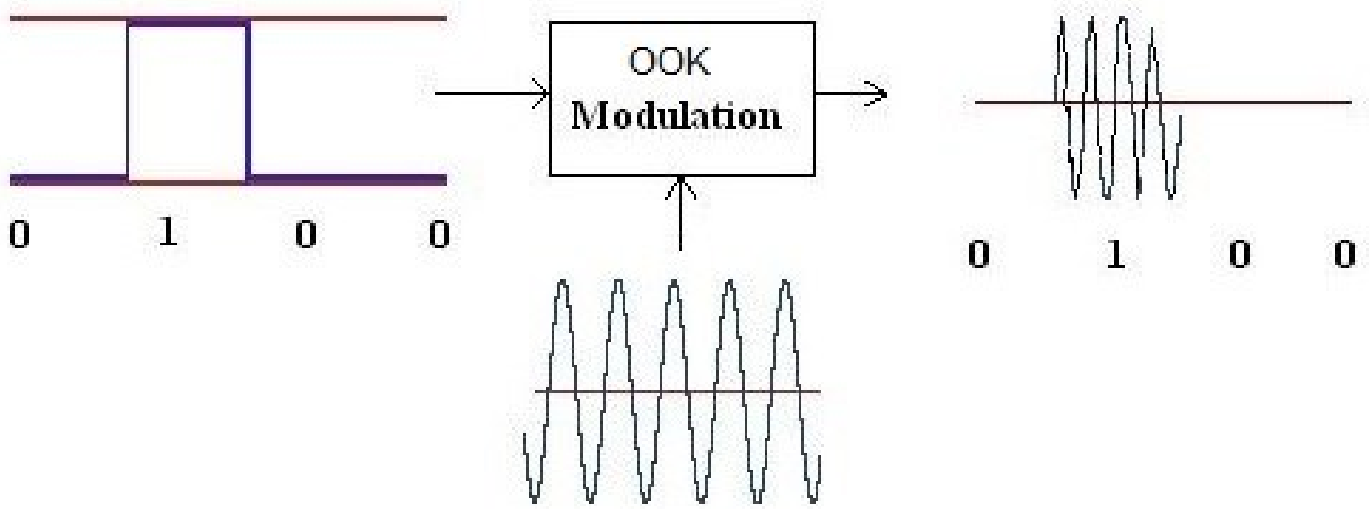
- High level changes of amplitude patterns of analog signals represent the data.
- Frequency of the analog signal remain the same throughout.

Frequency remains the same as carrier signal.



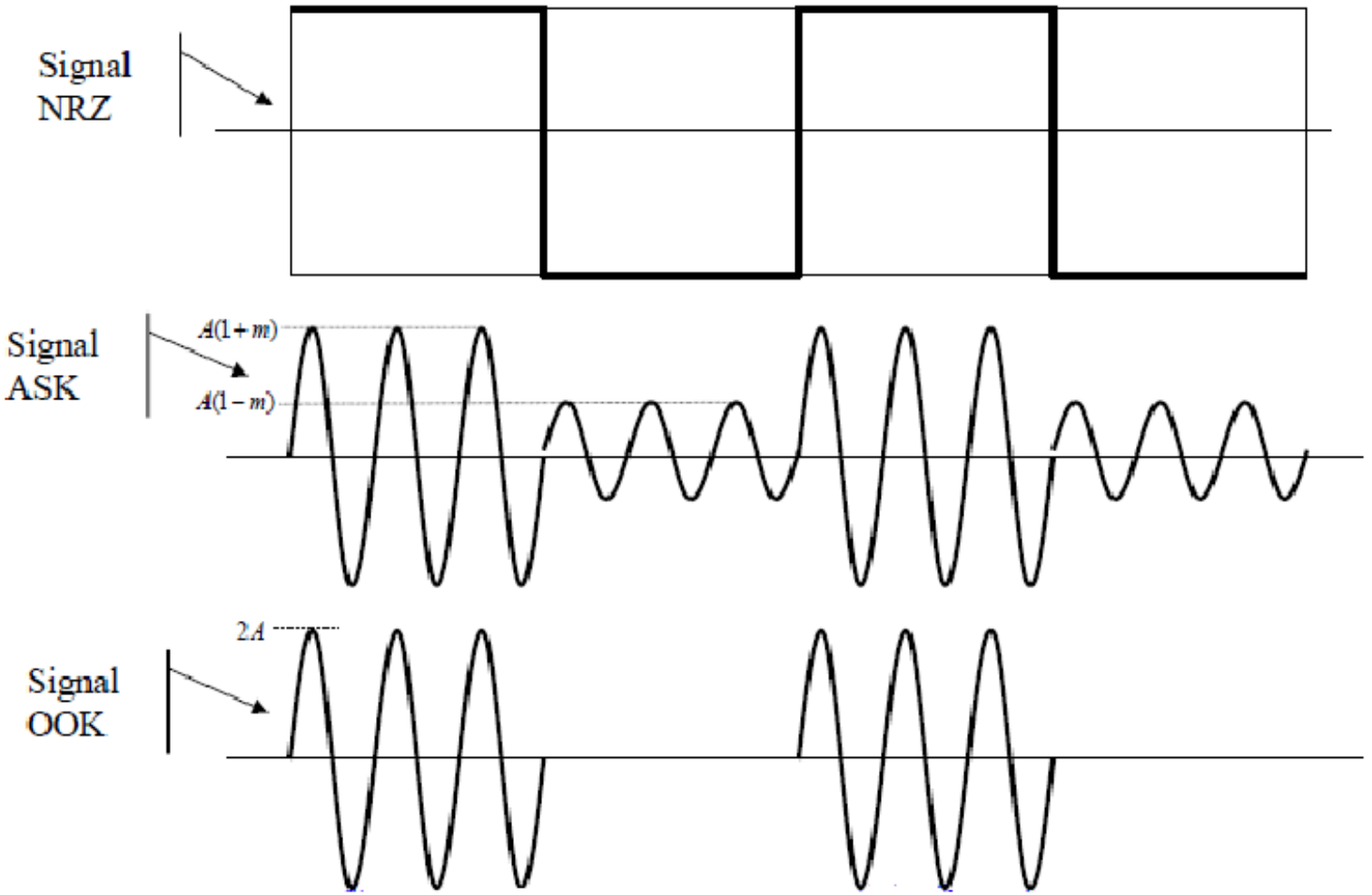
Amplitude Shift Keying (ASK)

- Noise (heat, electromagnetic interference, and induced noise) usually affects the amplitude of the signal, frequency is not really affected.
- Normally **binary** amplitude shift keying is used.
- A popular ASK technique is called on/off keying (OOK).
  - **One of the bit values** is represented by **no voltage**.
  - This reduce the amount of energy requirement.



Amplitude Shift Keying (ASK)

- The shifts in amplitude determines the symbols.



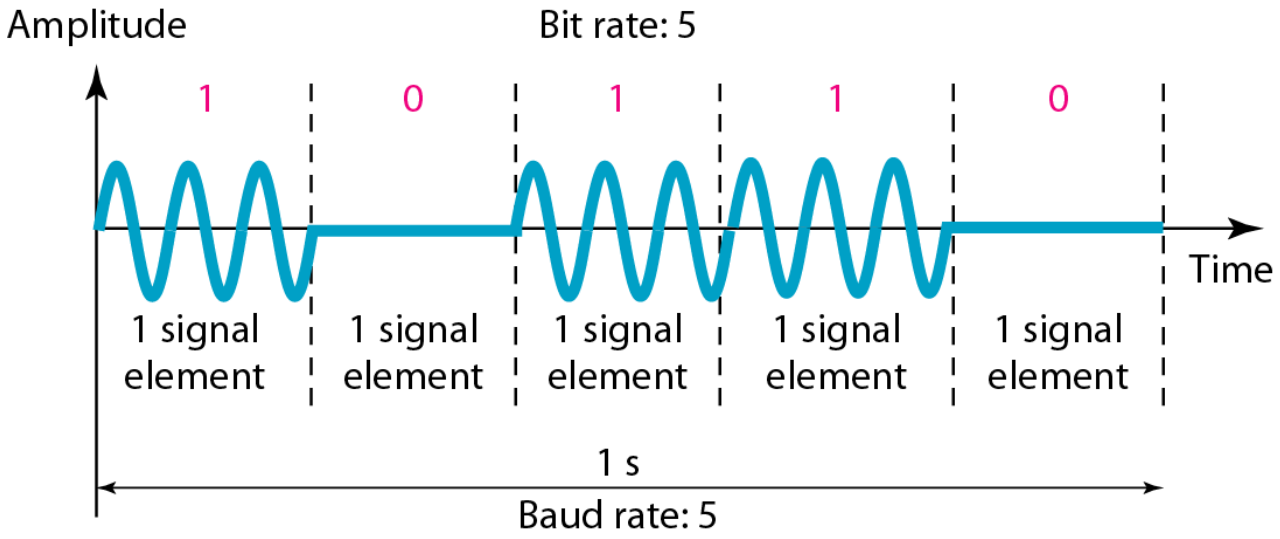
Amplitude Shift Keying (ASK)

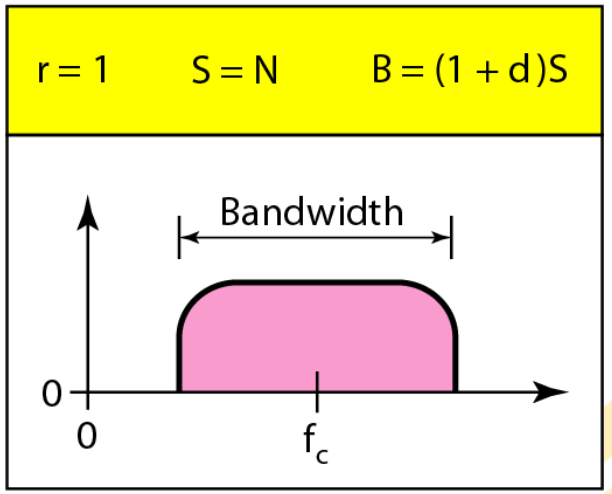
- **In ASK the baud rate is the same as the bandwidth** The bandwidth of the ASK signals is centered around a carrier frequency. So, we have

$$f_{MIN} = f_C - f_p = f_C - \frac{1}{2T_s} = f_C - \frac{N_{baud}}{2}$$

$$f_{MAX} = f_C + f_p = f_C + \frac{1}{2T_s} = f_C + \frac{N_{baud}}{2}$$

S = signal rate
N = bit rate
r = bit to signal ratio
d = modulation factor





ASK Example (Half-duplex)

- Given a bandwidth of 5000 Hz for an ASK signal, what are the baud rate and bit rate? (Note: d=0, bandwidth=S=N)
- In ASK the baud rate is the same as the bandwidth,
- so the baud rate is
- But because the baud rate and the bit rate are also the same for ASK,
- So, the bit rate is

ASK Exercise 1 (Half-Duplex)

- We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz.
- What are the carrier frequency and the bit rate if we modulated our data by using binary ASK with d = 1?

- The middle of the bandwidth is located at ☐ kHz.
- This means that our carrier frequency can be at ☐ kHz.
- We can use the formula for bandwidth to find the bit rate (with d = 1 and r = 1).

- We can calculate the bit rate N as follows

$$BW = (1 + d)S$$

$$100\ \text{kHz} = (1 + 1)S$$

$$S = \square\ kbaud;\ N = \square\ kbps$$

ASK Exercise 2 (Half-duplex)

- We have an available bandwidth of 200 kHz which spans from 400 to 600 kHz.
- What are the carrier frequency and the bit rate if we modulated our data by using binary ASK with d = 1?

- This means that our carrier frequency can be at ☐ kHz.
- We can use the formula for bandwidth to find the bit rate (with d = 1 and r = 1).

- We can calculate the bit rate N as follows

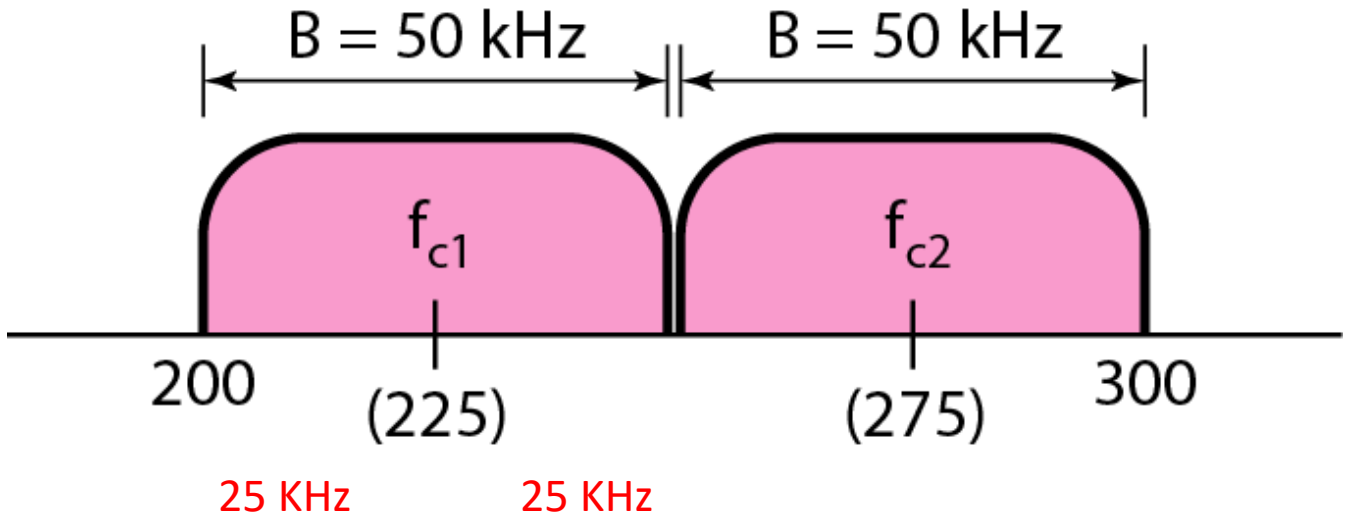
$$BW = (1 + d)S$$

$$\square = (1 + 1)S$$

$$S = \square \; kbaud; \; N = \square \; kbps$$

Full-Duplex ASK Example 1

- In data communications, we normally use full-duplex links with communication in both directions. We need to divide the bandwidth into two with two carrier frequencies, as shown in Figure below.
- The figure shows the positions of two carrier frequencies and the bandwidths. The available bandwidth for each direction is now 50 kHz, which leaves us with a data rate of 25 kbps in each direction.



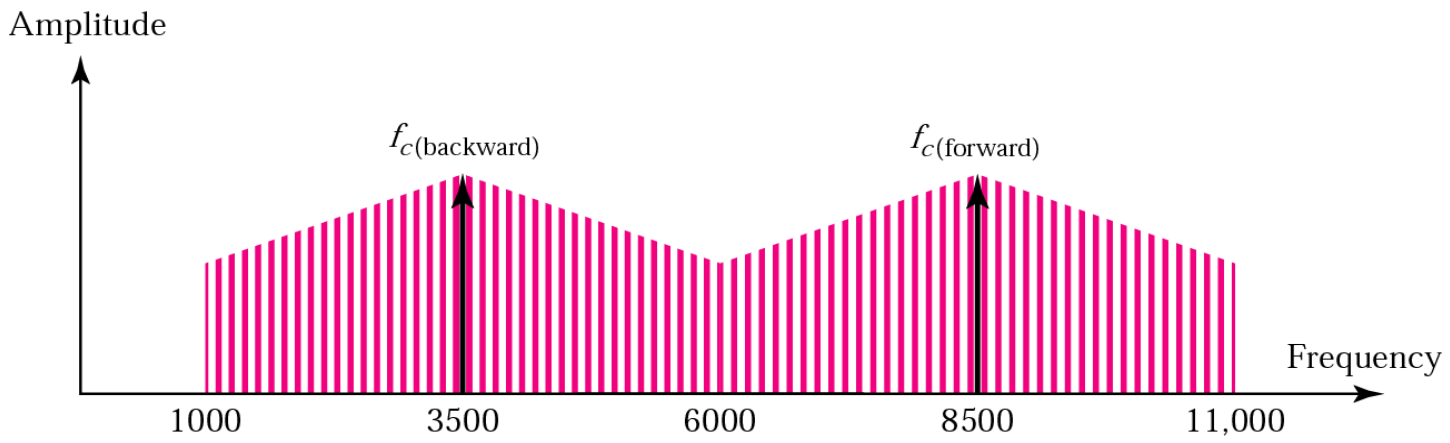
Full-Duplex ASK Example 2

- Given a bandwidth of 10,000 Hz (1000 to 11,000 Hz), draw the full-duplex ASK diagram of the system. Find the carriers and the bandwidths in each direction. Assume there is no gap between the bands in the two directions.

- For full-duplex ASK, the bandwidth is divided equally for each direction

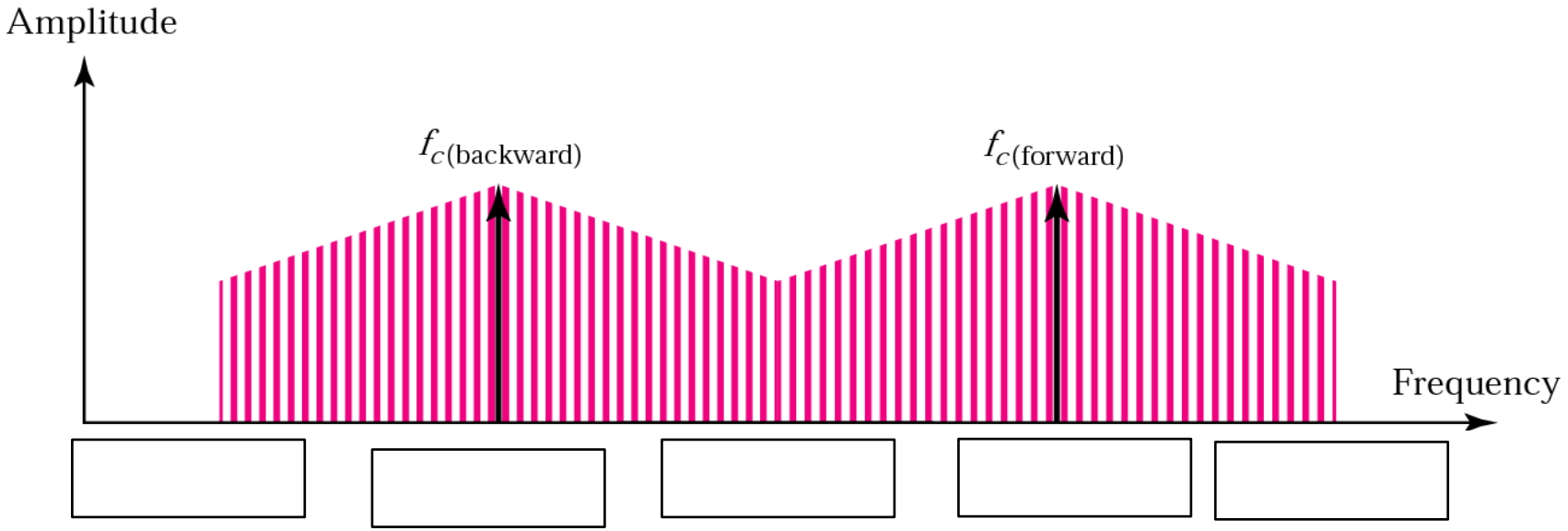
$$BW = \frac{10000}{2} = 5000\ Hz$$

- The carrier frequencies can be chosen at the middle of each band. We have



Full-duplex ASK Exercise 1

- Given a bandwidth of 4,000 Hz (1,000 to 5,000 Hz), draw the full-duplex ASK diagram of the system. Find the carriers and the bandwidths in each direction. Assume there is no gap between the bands in the two directions.



Do exercises
Let's have a Break !



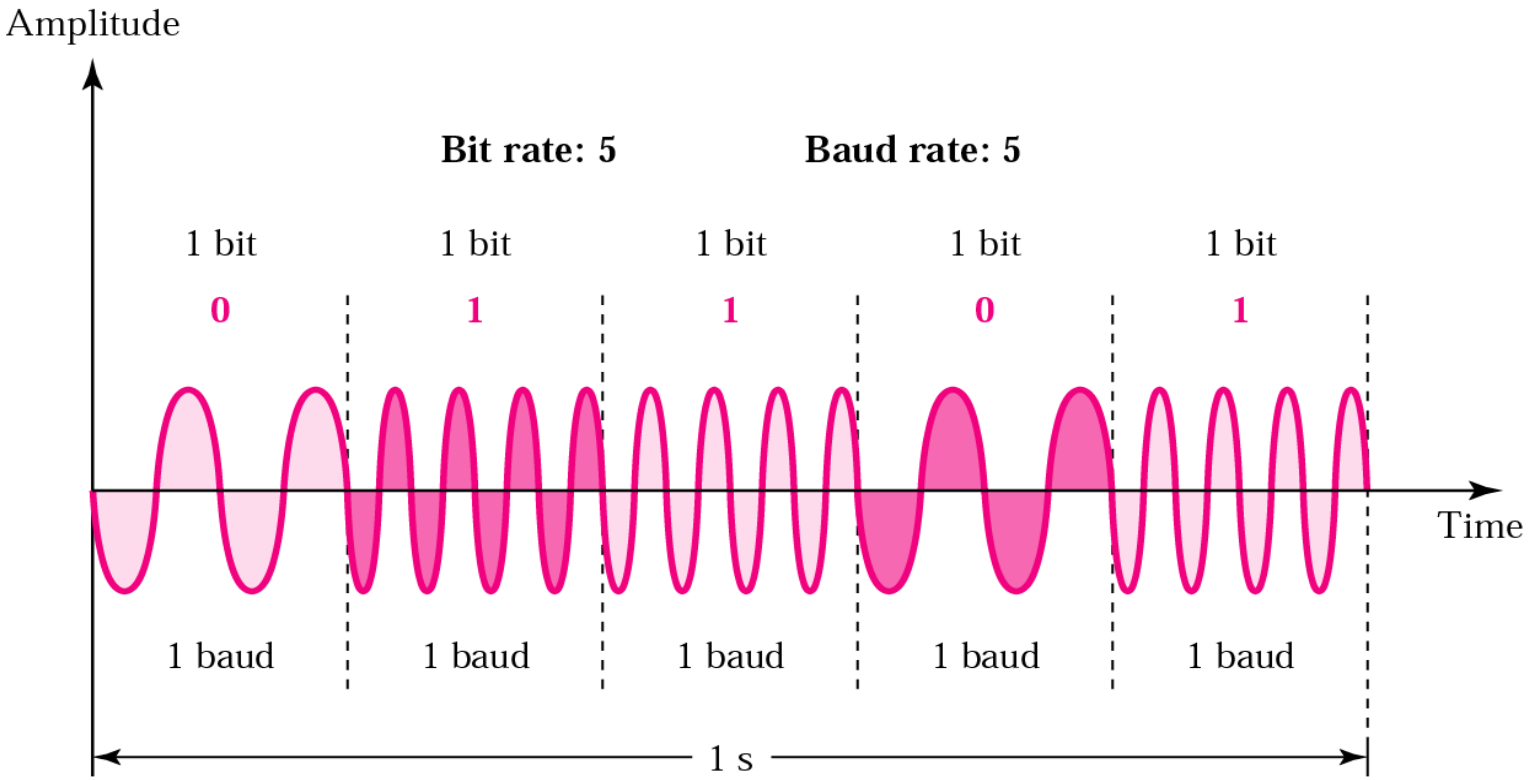






Frequency Shift Keying (FSK)

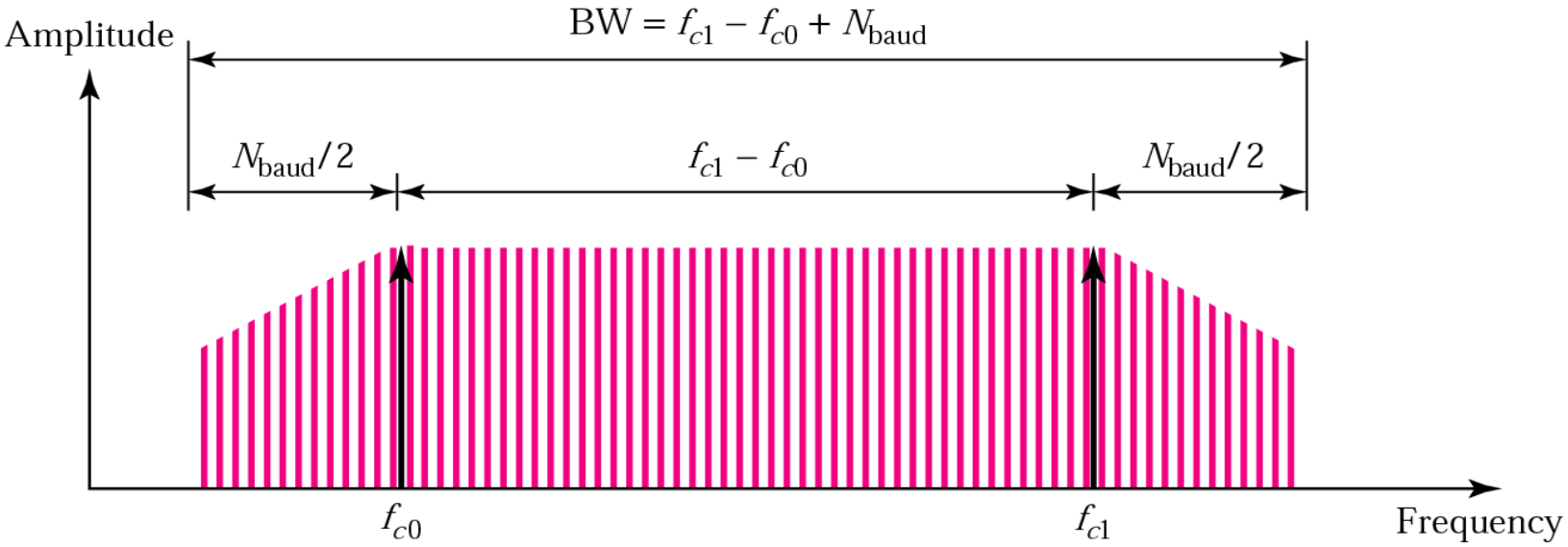
- Frequency is normally not affected by noise. We can modulate the frequency of the signal to digital values.
- Two or more carrier signals are used.



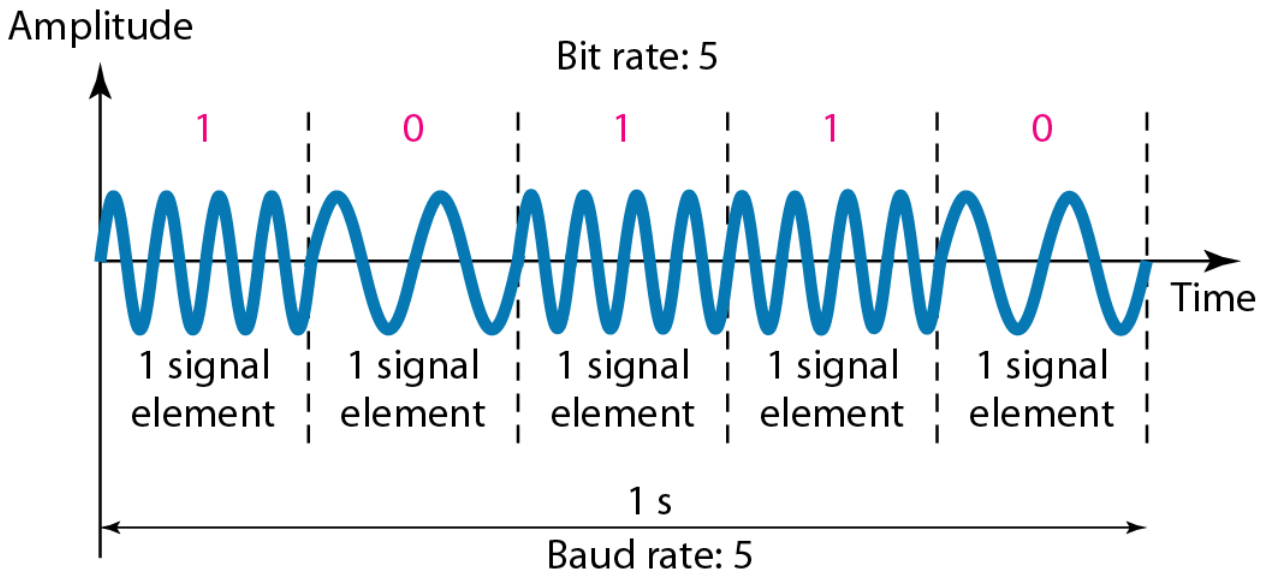
Frequency Shift Keying (FSK)

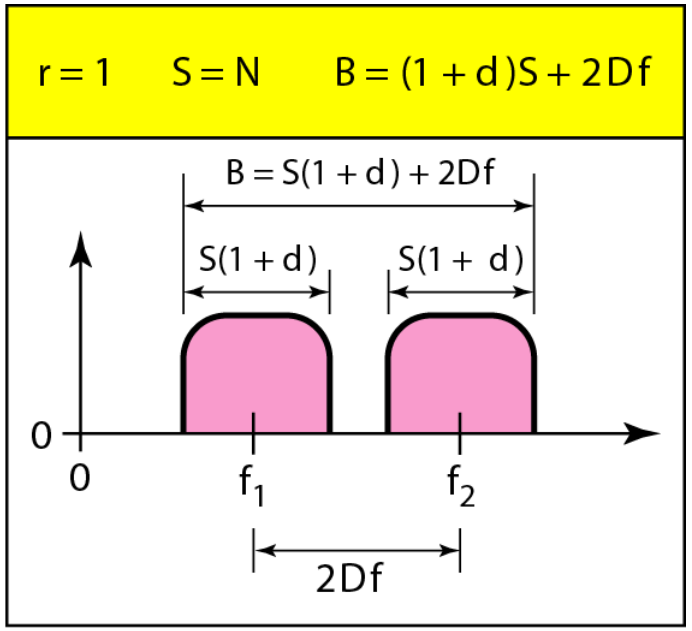
- Bandwidth: in case that we use 2 carrier signals with 2 frequencies

$$BW = f_{c1} - f_{c0} + N_{baud}$$



Frequency Shift Keying (FSK)





Frequency Shift Keying (FSK)

- Find the minimum bandwidth for an FSK signal transmitting at 2000 bps. Transmission is in half-duplex mode, and the carriers are separated by 3000 Hz ($f_{c1} - f_{c0}$).

$$BW = f_{c1} - f_{c0} + N_{baud} = \boxed{}$$

Nbaud=S=N=2000

FSK Example 1

- Find the maximum bit rates for an FSK signal if the bandwidth of the medium is 12,000 Hz and the difference between the two carriers is 2000 Hz. Transmission is in full-duplex mode.

- Because the transmission is full duplex, only **6000 Hz** is allocated for each direction.

$$BW = f_{c1} - f_{c0} + N_{baud}$$

$$N_{baud} = \boldsymbol{BW} - (f_{c1} - f_{c0}) = \boxed{}$$

- Since r = 1 and the baud rate is the same as the bit rate.

- In this example, the bit rate is $\boxed{}$

r = 1	S = N	B = (1 + d)S + 2Df

FSK Exercise 1

- Find the maximum bit rates for an FSK signal if the bandwidth of the medium is 36,000 Hz and the difference between the two carriers is 10,000 Hz. Transmission is in full-duplex mode.

- Because the transmission is full duplex, only [ ] Hz is allocated for each direction.

$$BW = f_{c1} - f_{c0} + N_{baud}$$

$$N_{baud} = BW - (f_{c1} - f_{c0}) = \boxed{}$$

- Since r = 1 and the baud rate is the same as the bit rate.

- In this example, the bit rate is. [ ]

r = 1	S = N	B = (1 + d)S + 2Df

FSK Exercise 2

- If the bandwidth of the medium is 10,000 Hz and the lower carrier frequency is 5,000 Hz, the bit rate is 2000 bps. Transmission is in full-duplex mode.
- What is the upper carrier frequency?
- Because the transmission is full duplex, only Hz is allocated for each direction.

$$BW = f_{c1} - f_{c0} + N_{baud}$$

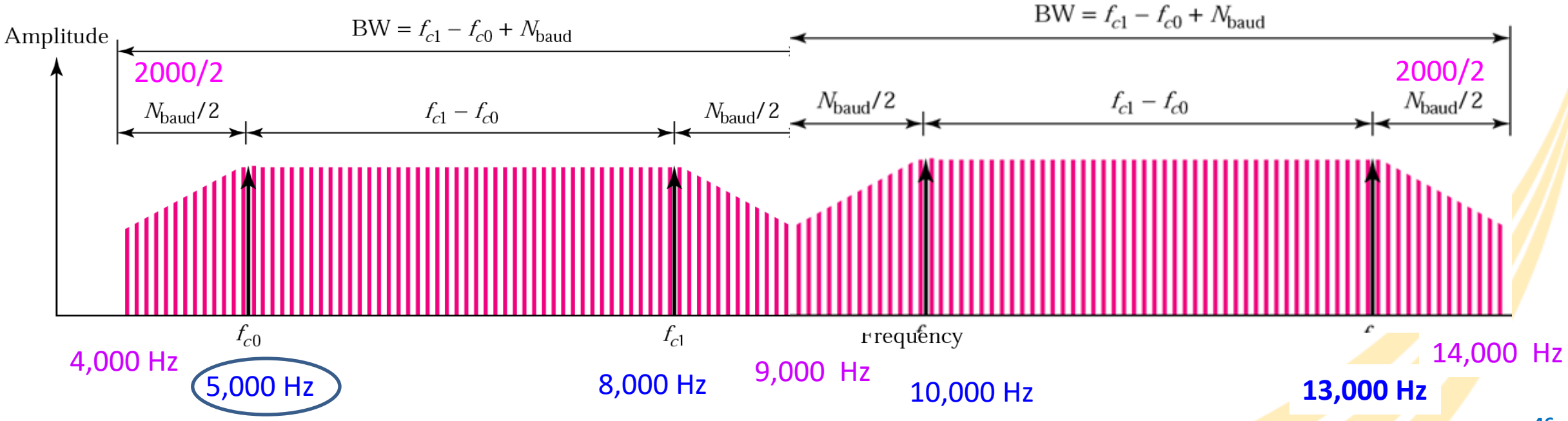
- Answer:
-

FSK Exercise 2 (Solution)

- Find the maximum bit rates for an FSK signal if the bandwidth of the medium is 10,000 Hz and the lower carrier frequency is 5,000 Hz, the bit rate is 2000 bps (Nbaud). Transmission is in full-duplex mode. What is the upper carrier frequency?

- Because the transmission is full duplex, only 10,000/2 Hz is allocated for each direction.

BW = 5,000 = Fc1 – Fc0 + 2000 → Fc1-Fc0 = 3,000

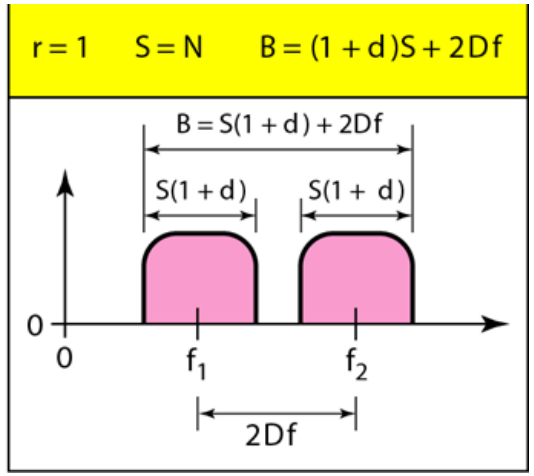


FSK Example 2

- We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz.
- What should be the carrier frequency and the bit rate if we modulated our data by using FSK with d = 1?

- The midpoint of the band is at 250 kHz. We choose **2Δf** to be **50 kHz**; this means

$$B = (1+d) \times S + 2\Delta f = 100 \quad \rightarrow \quad 2S = 50 \text{ kHz} \quad S = 25 \text{ kbaud} \quad N = 25 \text{ kbps}$$



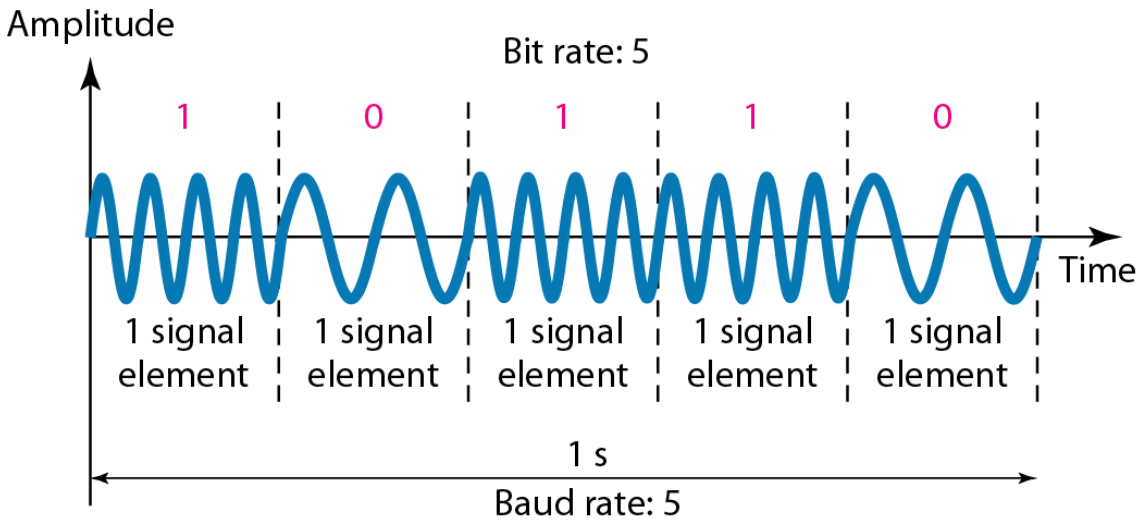
fc1 fc2

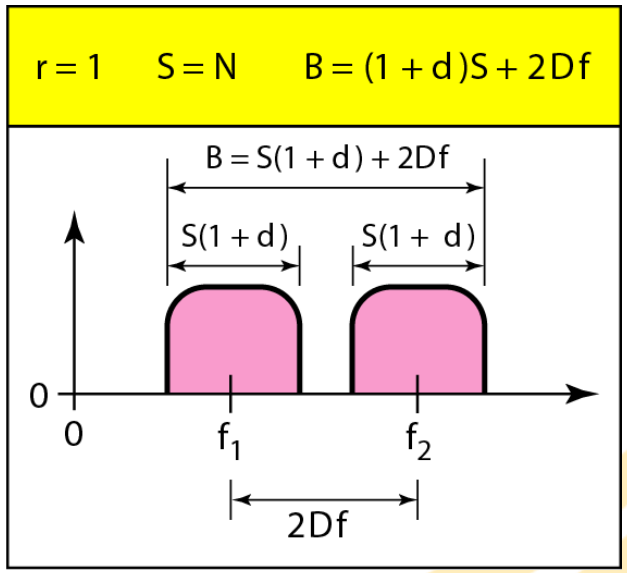
200 KHz 250 KHz 300 KHz

fc1-fc2=275-225=50

FSK Example 3

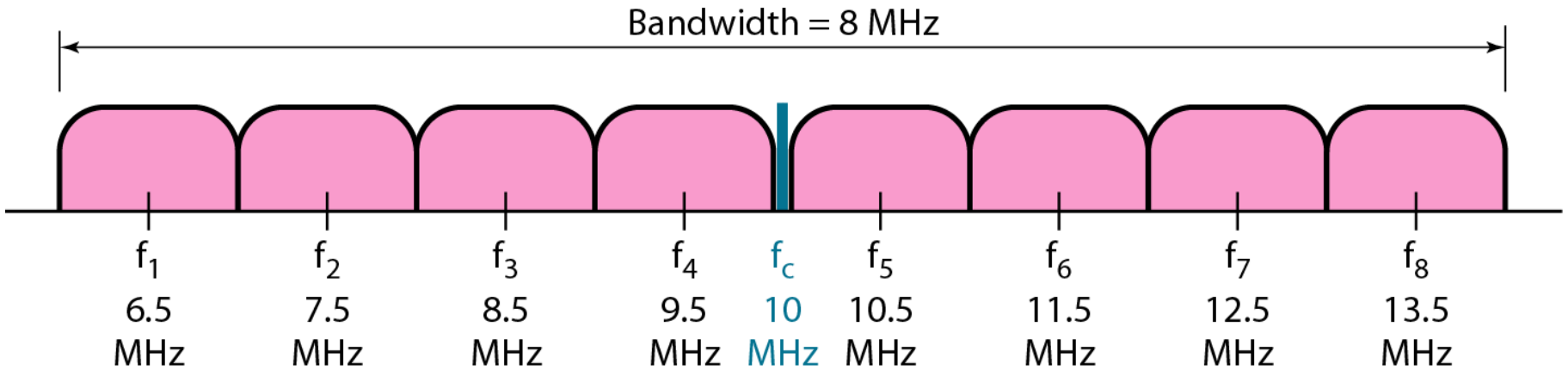
- We need to send data 3 bits at a time at a bit rate of 3 Mbps. The carrier frequency is 10 MHz. Calculate the number of levels (different frequencies), the baud rate, and the bandwidth.
- In this case, 3 bits per symbol. This means that the number of level (L) is $2^3 = 8$.
- The baud rate is therefore (S=N/r)=3 Mbps / 3 = 1 Mbaud per second. The bandwidth for each carrier is therefore 1 MHz.
- Recall: 2 levels, two frequencies, what if we have 8 levels? 8 carrier frequencies





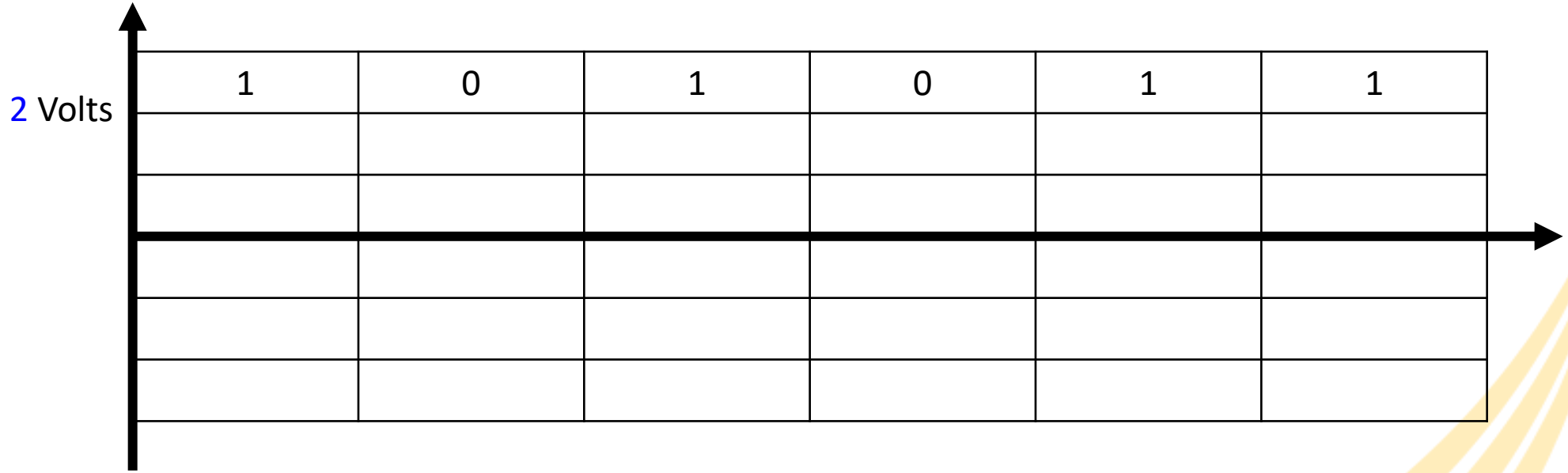
FSK Example

- Recall: 2 levels, two frequencies, what if we have 8 levels?
  - We should use 8 carrier signals.
  - Each signal has 1 MHz bandwidth.



FSK Exercise 3

- Draw the signal graph when the sender what to send data "**101011**" by using **FSK** encoding (bit 0 for 1 Hz and bit 1 for 2 Hz). Assume the amplitude is 2 volts, and Phase is 0 degree.
- Note: You might use a device such as tablet (iPad/Tab/writing Pads) to draw or PowerPoint draw tools.



You may assume bit 0 = 1 Hz (low frequency), bit 1 = 2 Hz (high frequency) and each slot of one bit is 1 sec. So, you may draw two cycles (two waves) for bit 1, and one cycle (one wave) for bit 0.

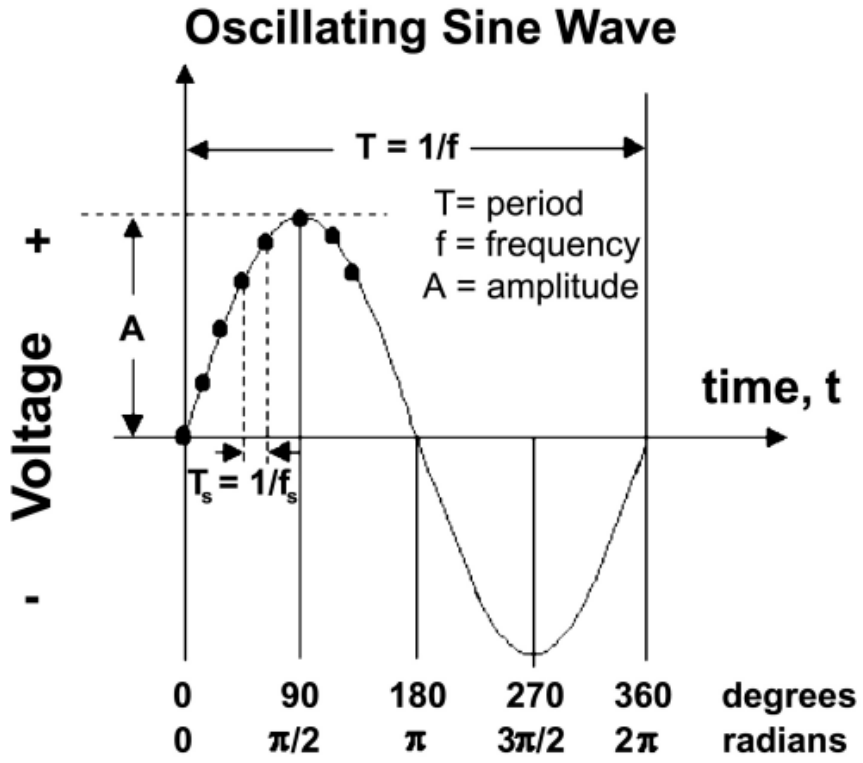
Efficiency Issues of ASK and FSK

- Must capture the amplitude or frequency information of the carrier
  - Require at least one cycle of a carrier wave to send a single bit
  - Transmission capacity is limited by the carrier frequency.
- A single sine wave circle
  - One amplitude
  - One frequency
  - Can be shifted with multiple phases!



Phase Shift Keying (PSK)

- Phase Shift Keying (PSK) is commonly used to send more bits during transmission.
  - PSK can change the phase of the carrier abruptly (phase shift)
  - Almost every real world modem use PSK.
- On a sine wave, each point correspond to a phase (angle)



See how sine wave can be created
https://www.youtube.com/watch?v=P-Umre5Np_0

Phase Shift Keying (PSK)

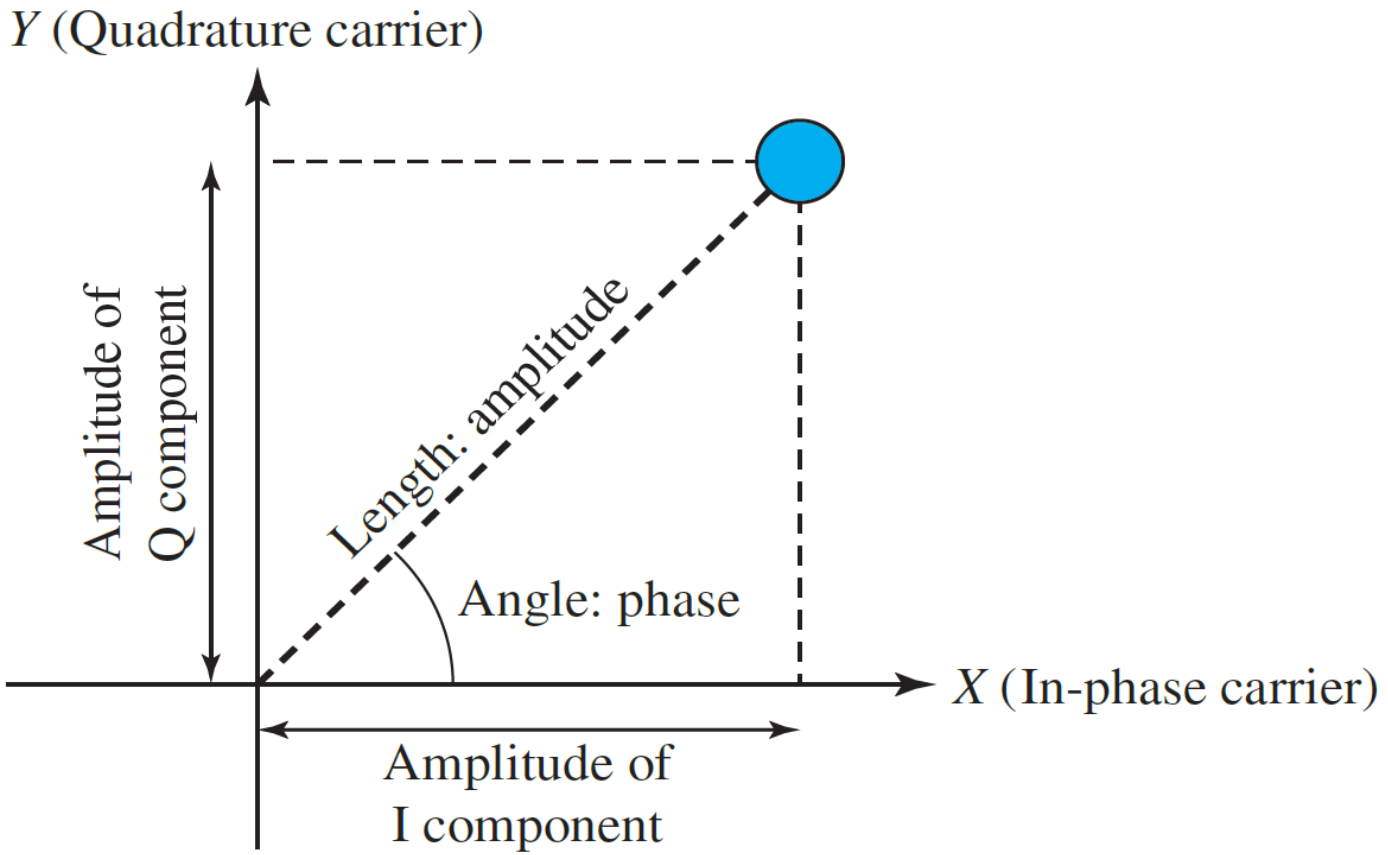


- We can modulate the phase of carrier signal.
- **PSK is not susceptible to the noise degradation as in ASK or to the bandwidth limitation of FSK.**
- Smaller variations in the signal can be detected by the receiver

- How to encode data into phase shifts?
  - A sender and receiver can agree on the number of bits per second.
  - A sender and receiver must agree on the phase shift values association with the data bits.
  - A **constellation diagram** is used to express the exact assignment of data bits to specific phase changes.

Phase Shift Keying (PSK)

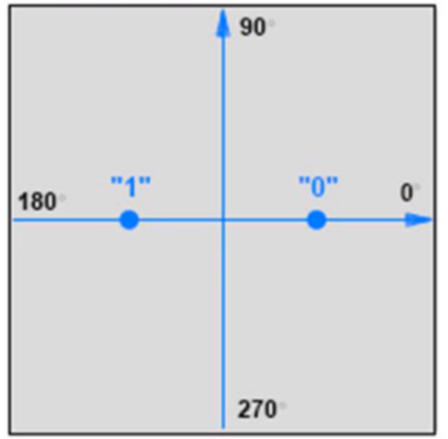
- Concept of constellation diagram



Phase Shift Keying (PSK)

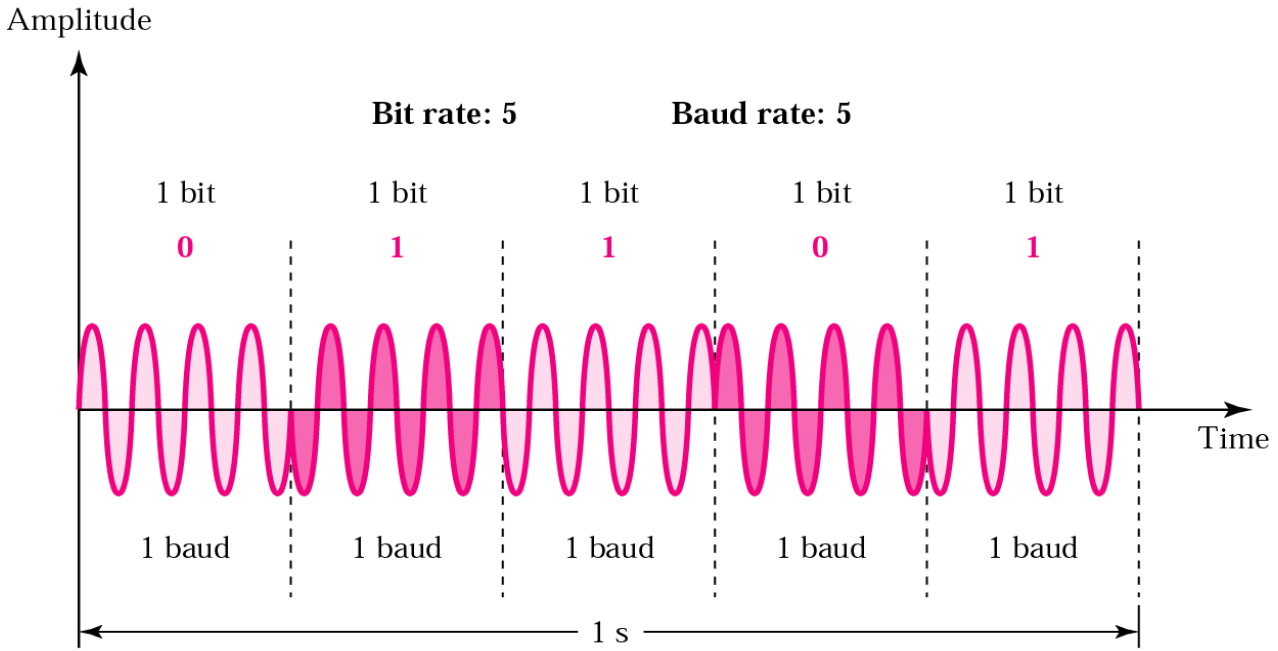
- **Constellation diagram**

0: 0 degree
1: 180 degree



1 bit / shift

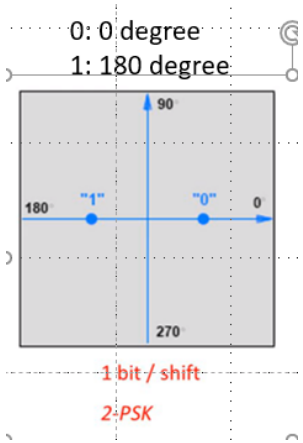
2-PSK

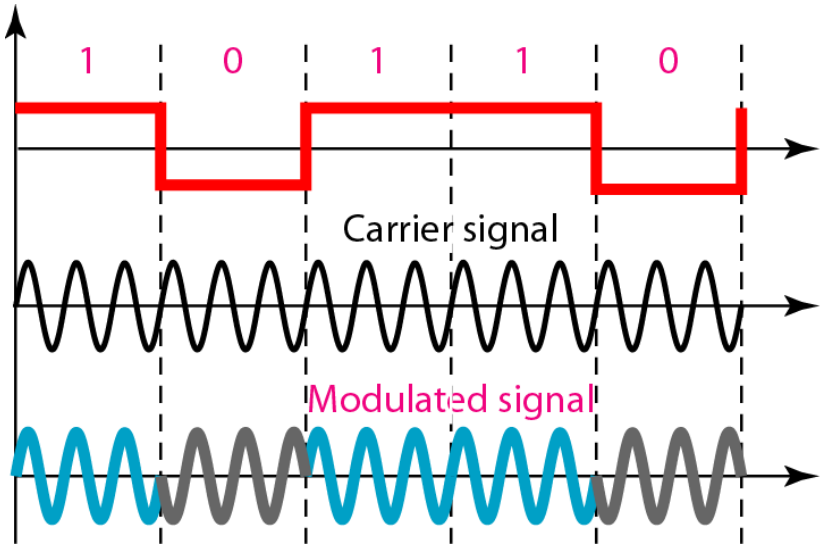


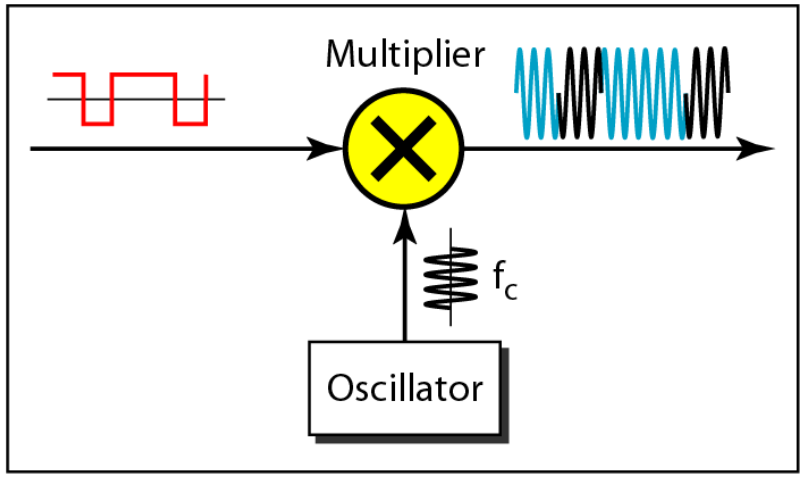
Phase Shift Keying (PSK)

- Assume we want to use binary PSK to send 4 bits "0101" in one second. Draw the resulting waveform

- Implementation of PSK

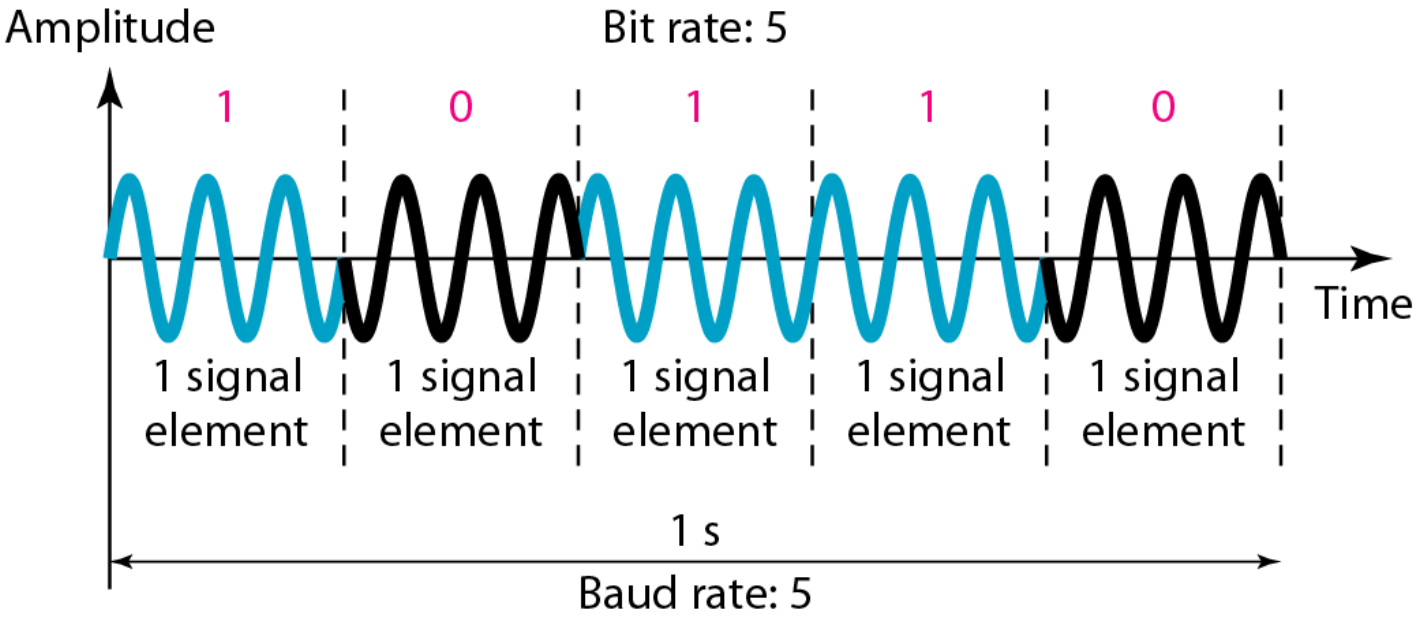


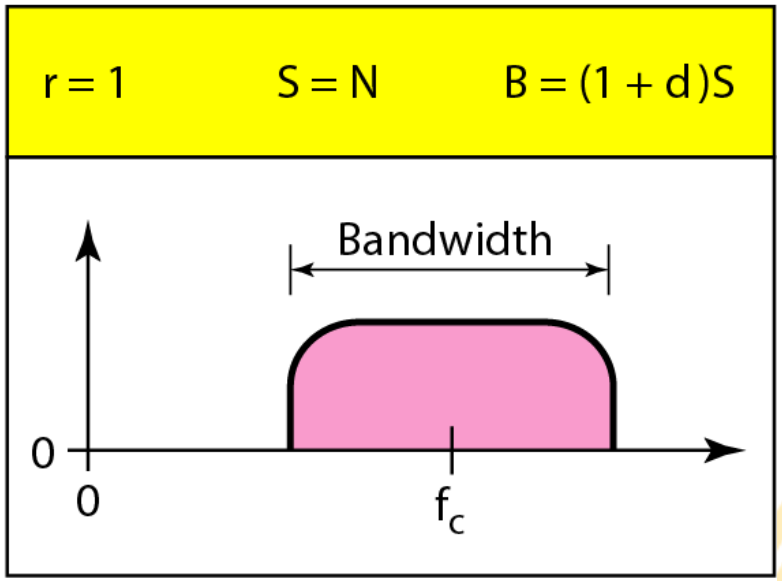




Phase Shift Keying (PSK)

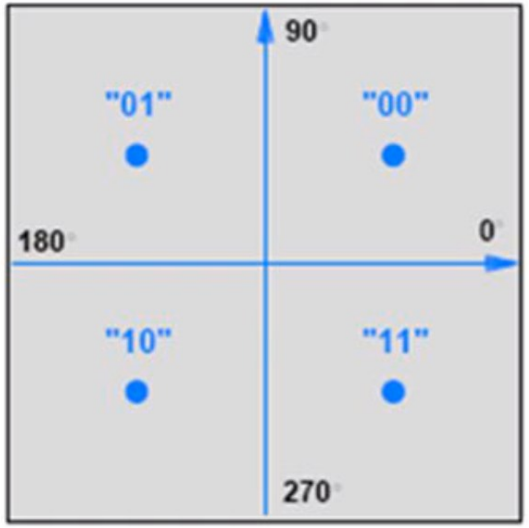
- Bandwidth requirements for binary PSK





Multiple-Level PSK

- We can create multiple phase shifts to send multiple bits at a time when there is a phase change. In this case, we are sending 2 bits at a time.



2 bit / shift

4-PSK

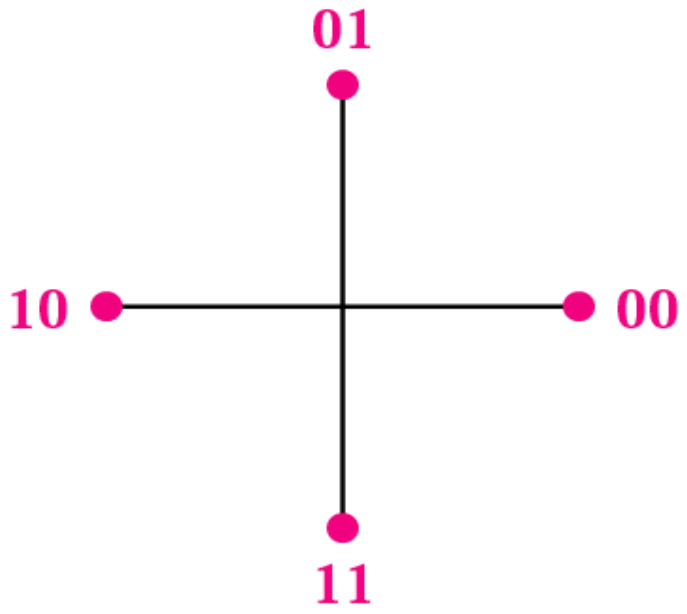
Phase Shift	45°	135°	225	315
Bit Value	00	01	10	11

Multiple-Level PSK

- There are many ways to designate the constellation

Dibit	Phase
00	0
01	90
10	180
11	270

Dibit
(2 bits)



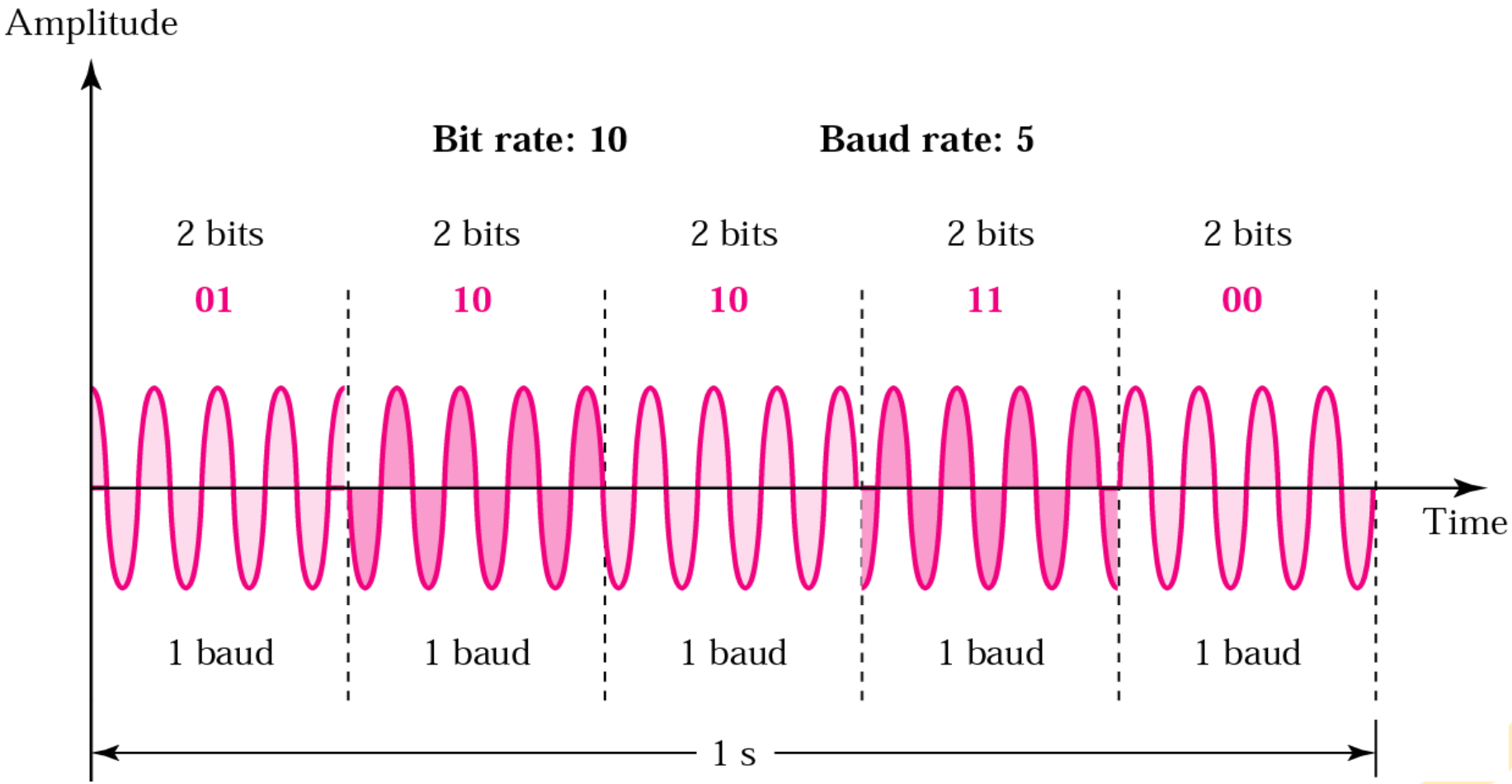
Constellation diagram

4-PSK or Q-PSK

- Therefore, it is possible to increase the data rate by decreasing the angular difference between phases.
- However, noise and distortion can affect this system.

Multiple-Level PSK

- 4-PSK or Q-PSK

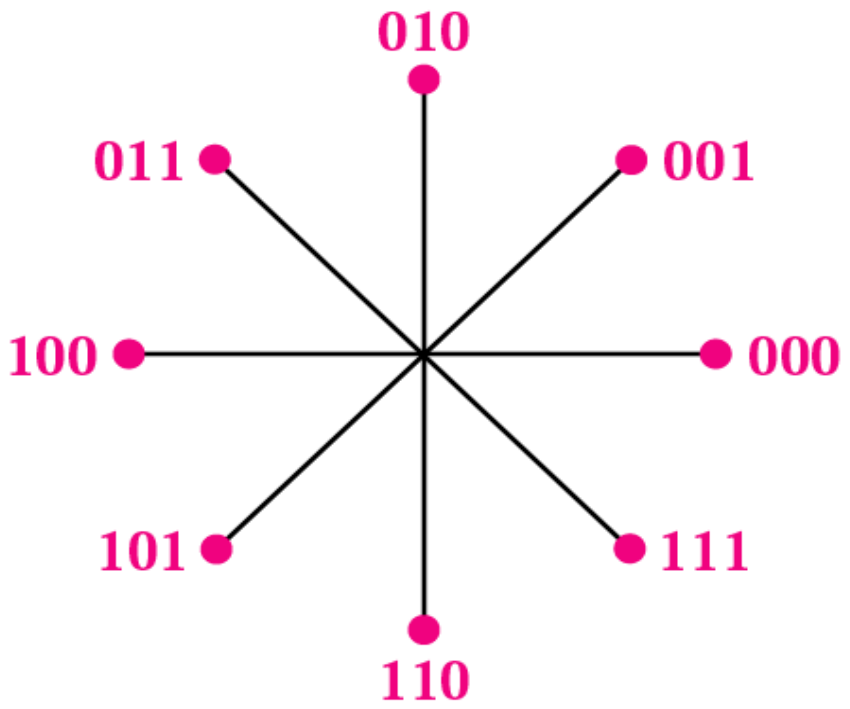


Multiple-Level PSK

- 8-PSK

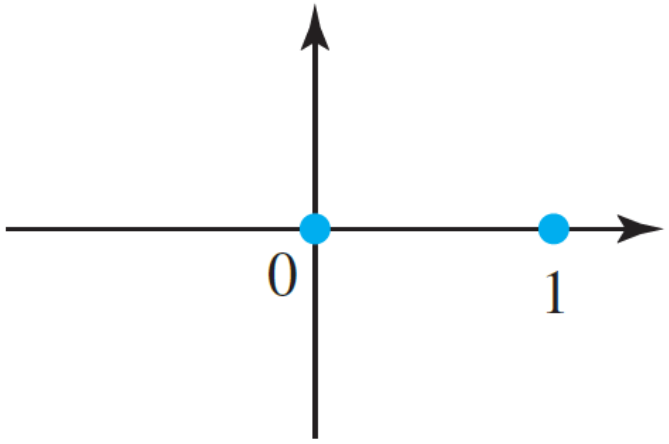
Tribit	Phase
000	0
001	45
010	90
011	135
100	180
101	225
110	270
111	315

Tribits
(3 bits)

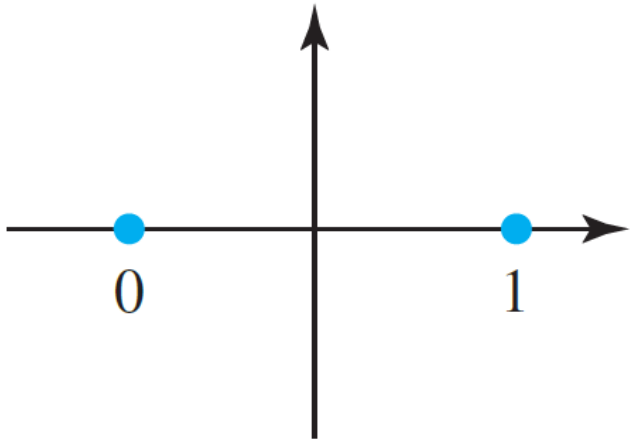


Constellation diagram

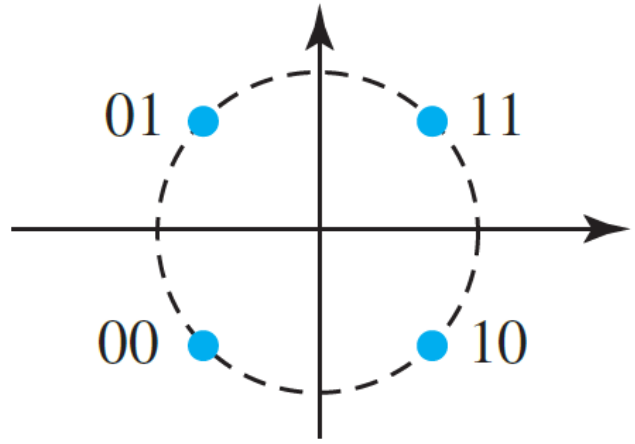
Constellation Diagrams



ASK (OOK)



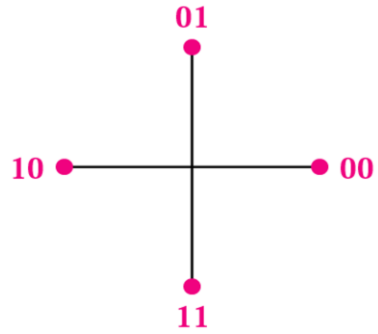
BPSK

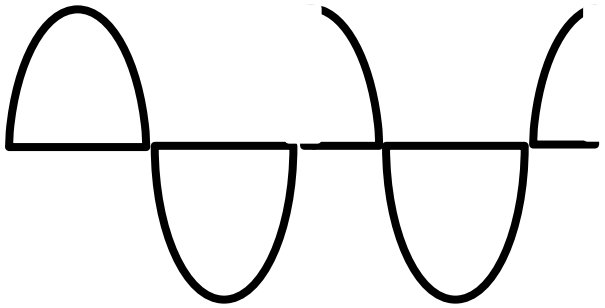


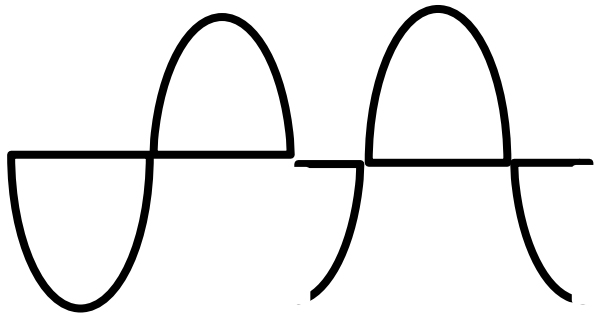
QPSK

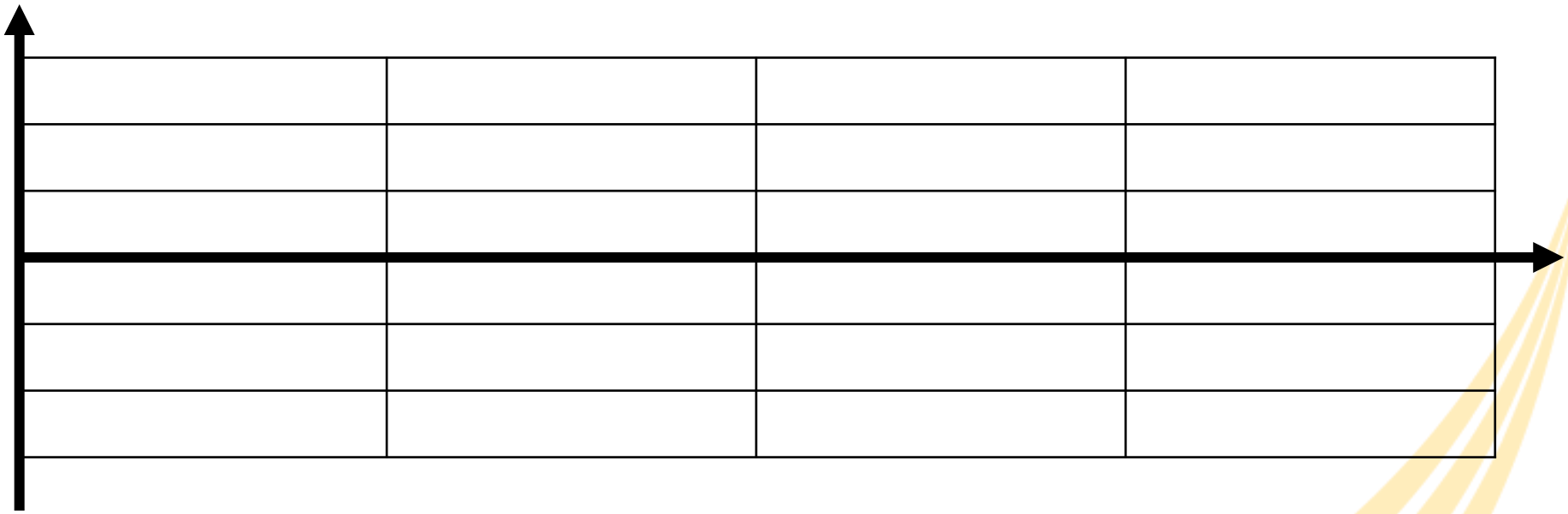
PSK Exercise

- Draw the signal graph when the sender what to send data "**10001101**" by using the following 4-PSK encoding.
- Assume Frequency = 1 Hz, Amplitude is 2 Volts





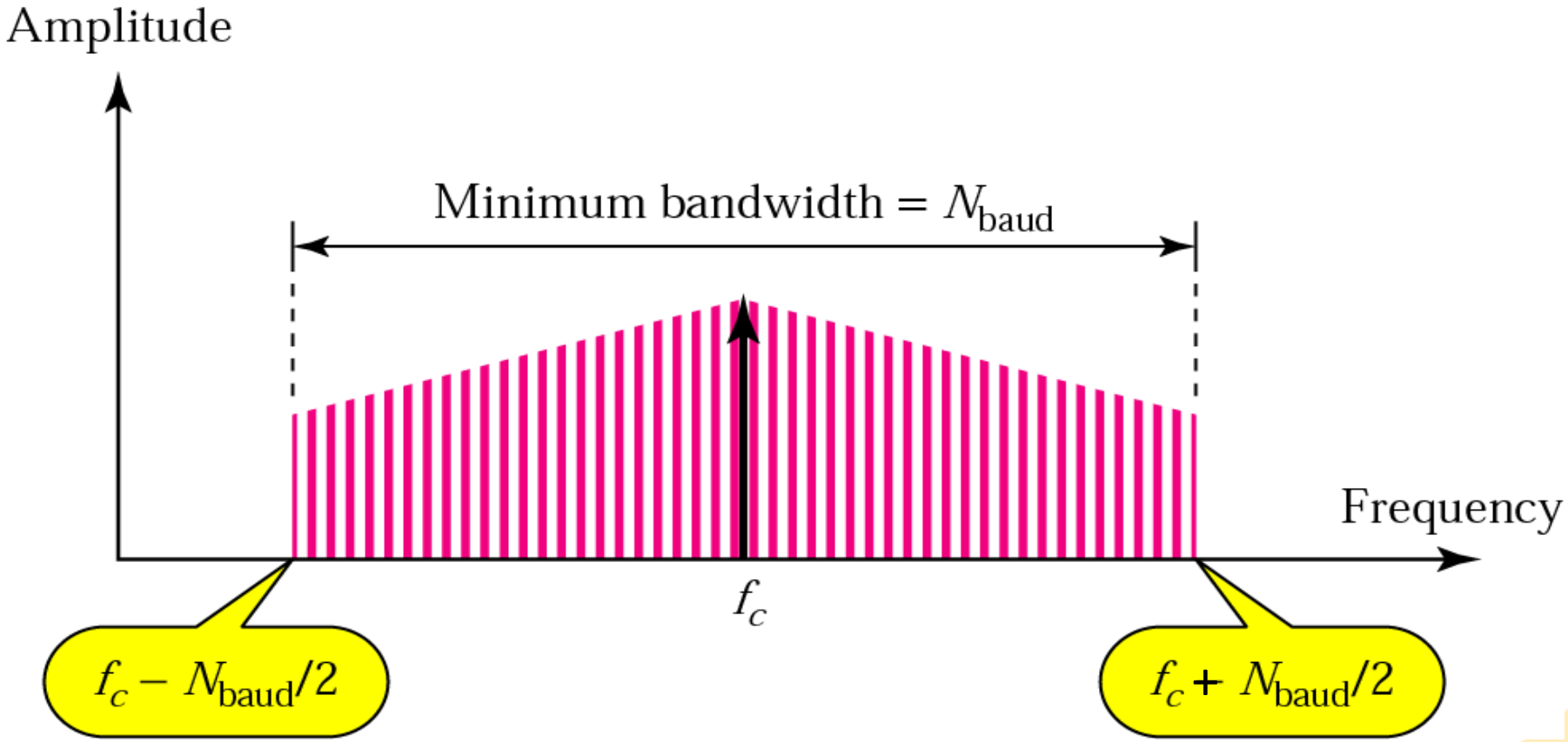




4-PSK can send 2 bits per one time. So, please grouping the data into two bits per each slot in your time-plot graph. Assume one slot in your graph is 1 sec as well. Therefore, 1 Hz = 1 cycle per one slot is enough.

PSK Bandwidth

- Multiple frequencies and phases of carrier signals are needed to create PSK in the way similar to ASK.



PSK Example 1

- Find the bandwidth for a B-PSK signal transmitting at 2000 bps. Transmission is in half-duplex mode.

- Solution:
  - In PSK the baud rate and bit rate are the same.
  - The baud rate is therefore
  - An PSK signal requires a minimum bandwidth equal to its baud rate.
  - Therefore, the minimum bandwidth is

PSK Example 2

- Given a bandwidth of 5000 Hz for an 8-PSK signal, what are the baud rate and bit rate?

- Solution:
  - For PSK the baud rate is the same as the bandwidth, which means the baud rate is 5000.
  - But in 8-PSK the bit rate is 3 times the baud rate, so the bit rate is

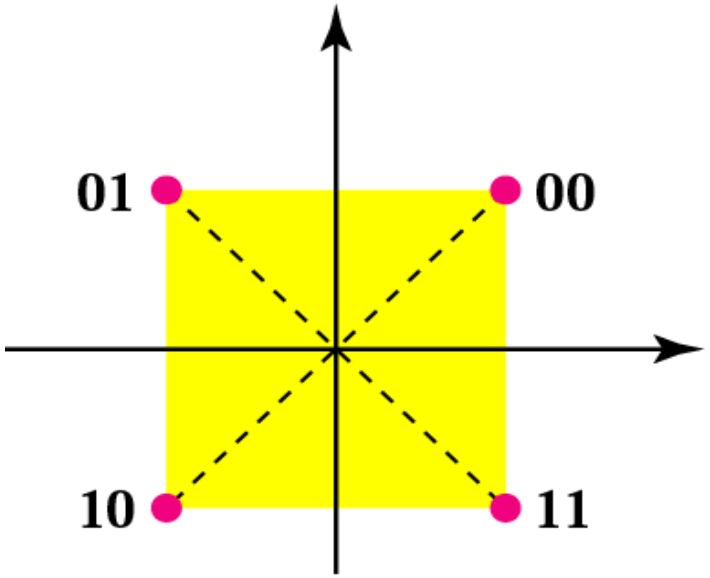
r=N/S = bit rate/baud rate = number of bits/number of signals
S*r = N

Quadrature Amplitude Modulation (QAM)

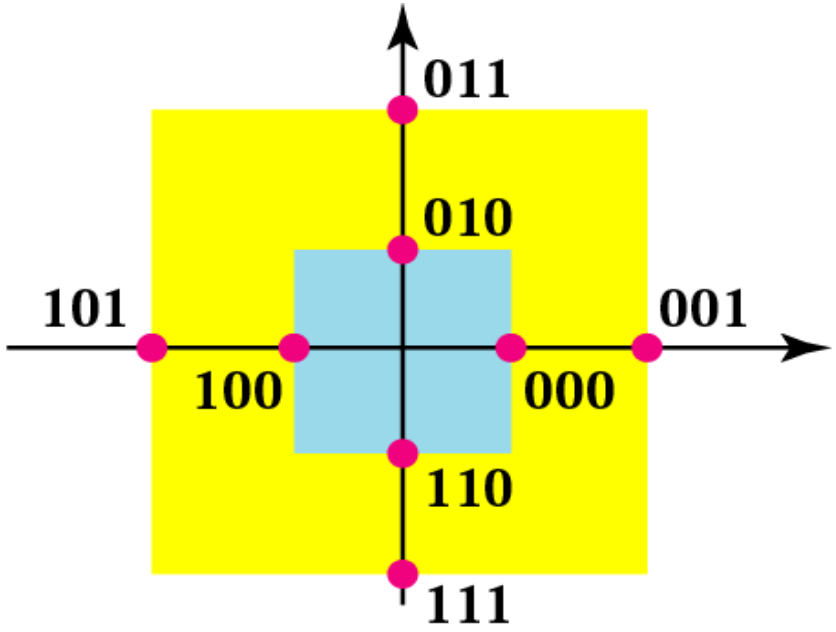
- PSK is limited by the ability of the equipment to distinguish small differences in phase
- Quadrature amplitude modulation is a combination of ASK and PSK so that a maximum contrast between each signal unit (bit, dibit, tribit, and so on) is achieved.
- We don't use FSK because FSK has **bandwidth limitation**.

The 4-QAM and 8-QAM (phase+amplitude)

- We don't need to get the phase change to be too small. We can also play with the change in the amplitude as well.



4-QAM
1 amplitude, 4 phases

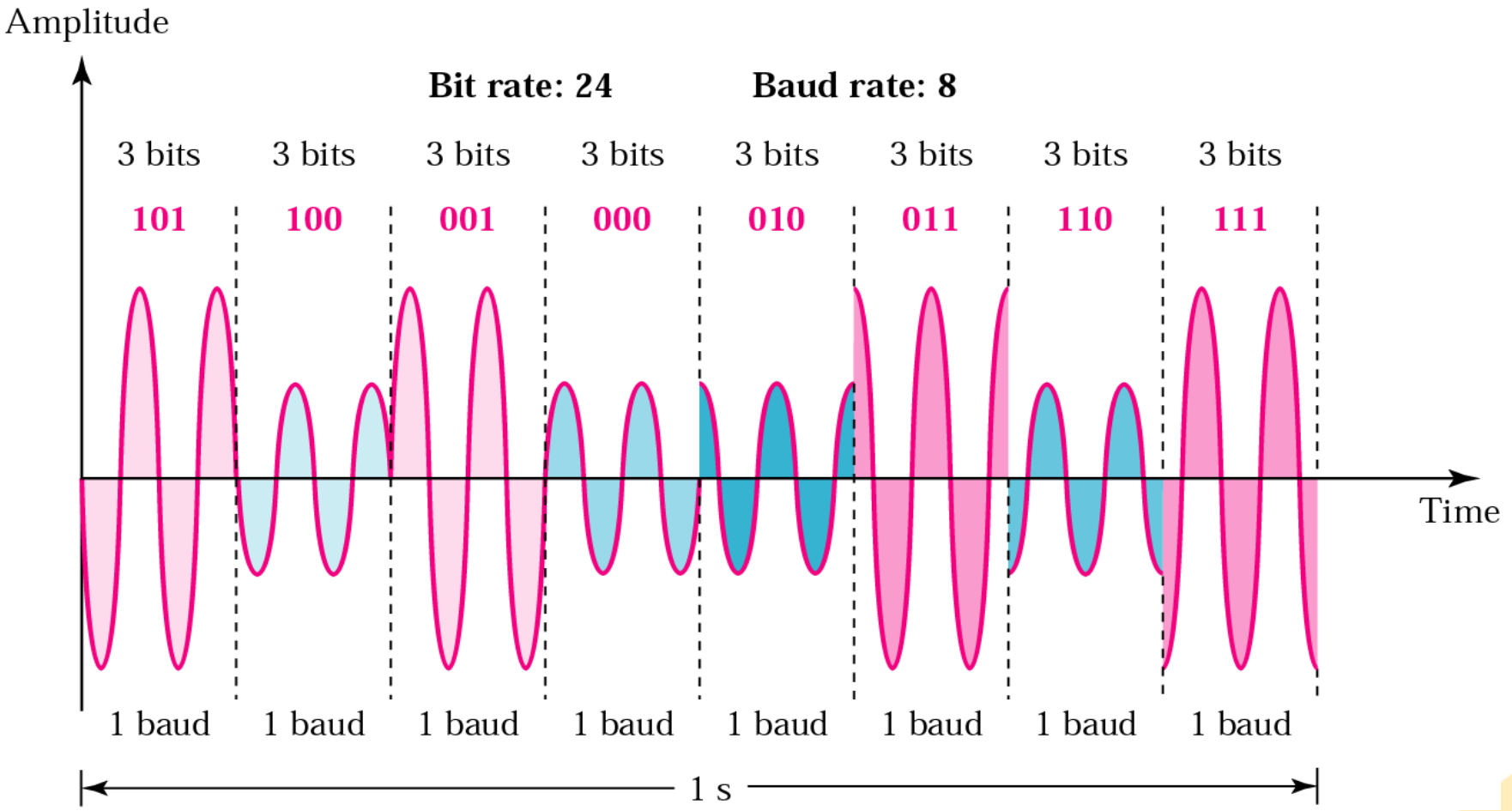


8-QAM
2 amplitudes, 4 phases

4 amp, 2 phases

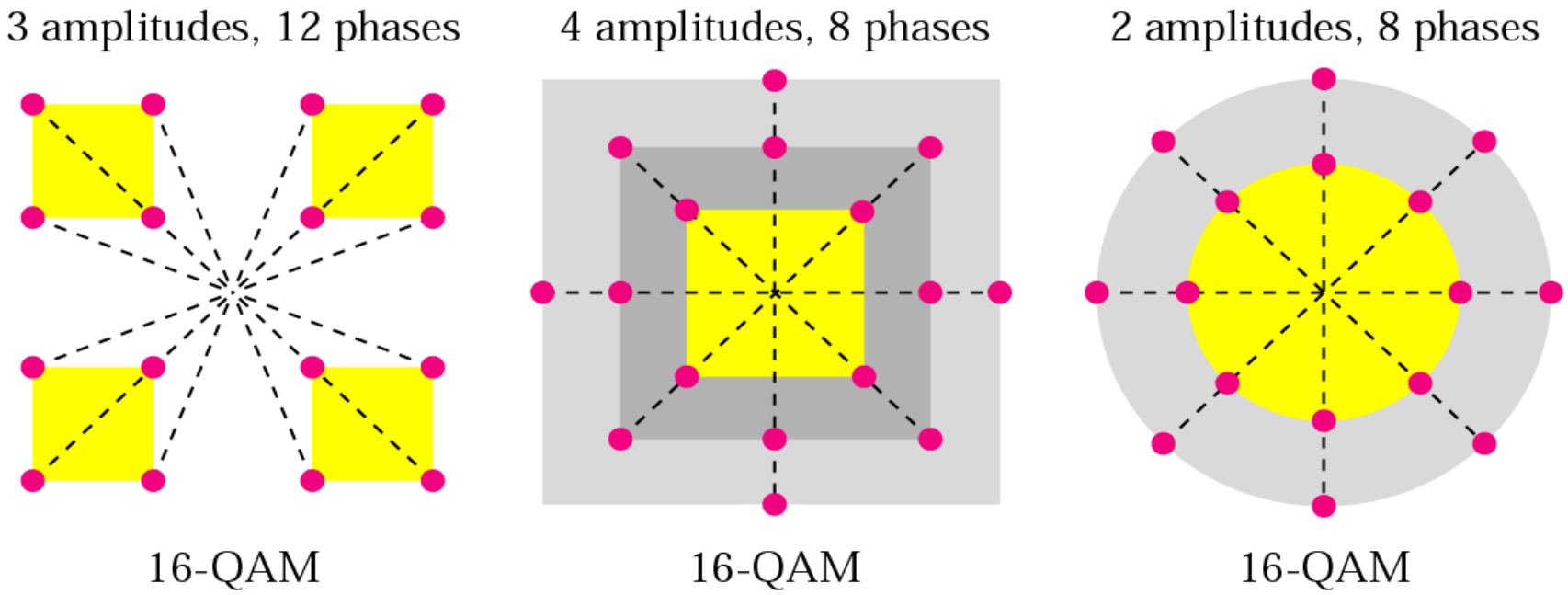
8-QAM (3-bit/signal)

- Time domain plot of the signals

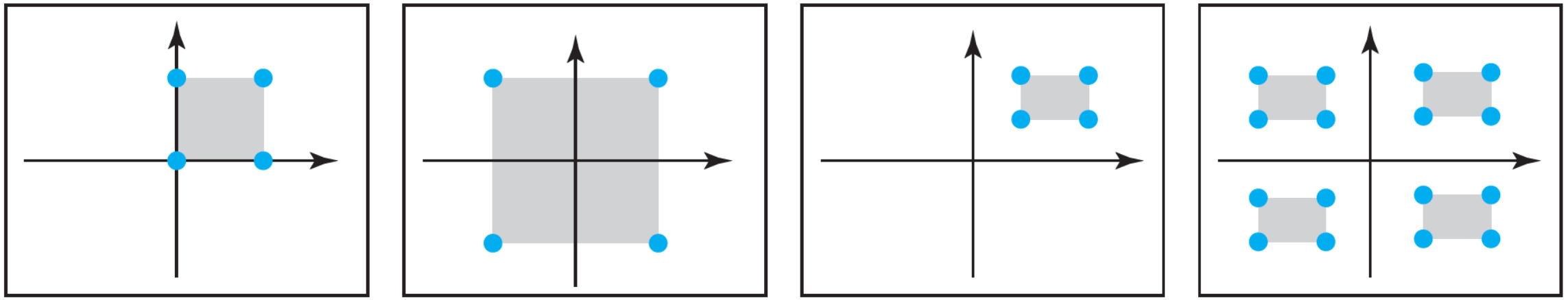


16-QAM (4-bit/signal)

- We can achieve better transmission rate with 16-QAM



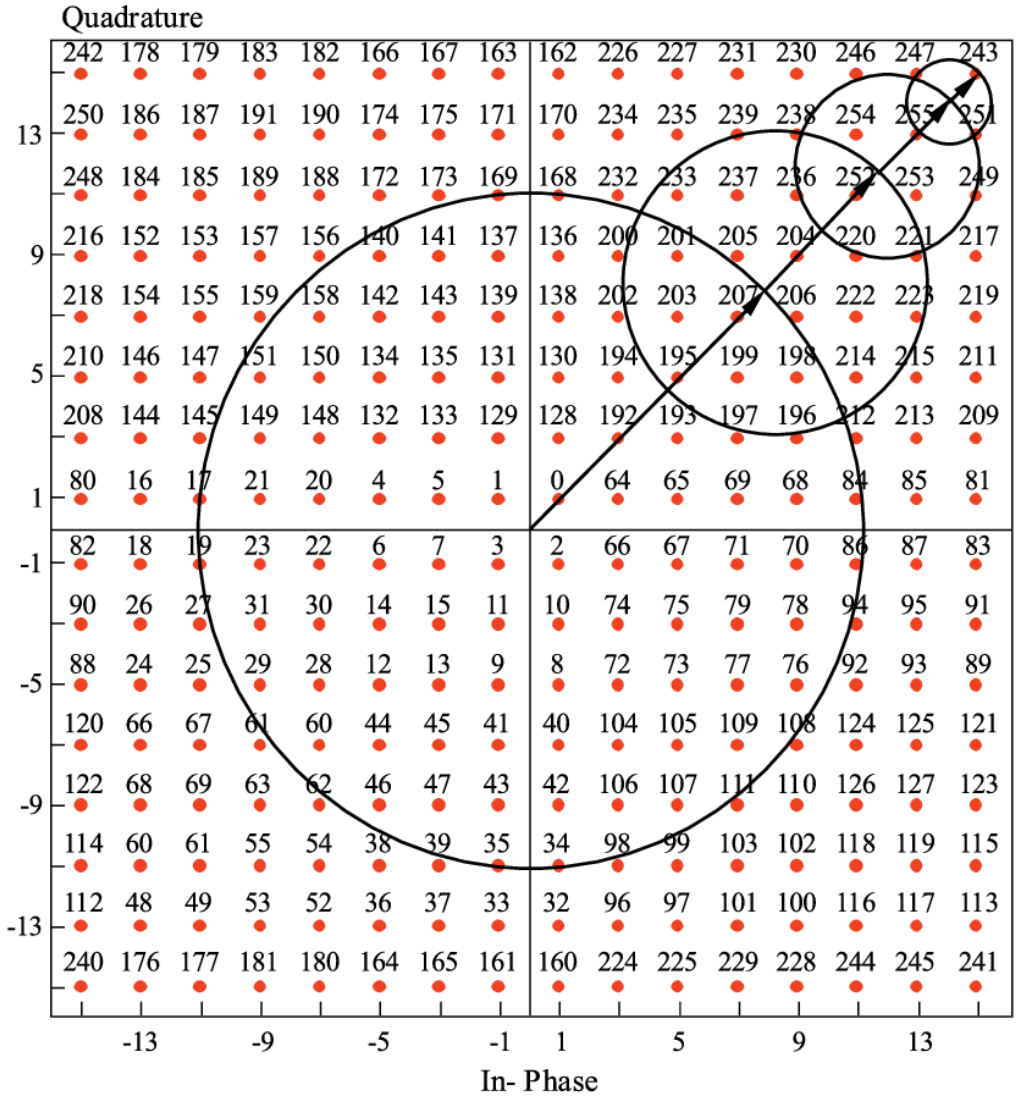
Constellation Diagrams

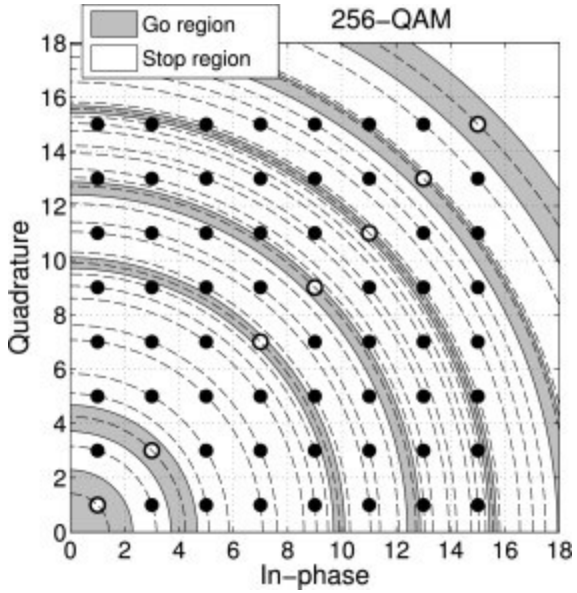


a. 4-QAM b. 4-QAM c. 4-QAM d. 16-QAM

256-QAM (8-bit/signal)

- 256-QAM is commonly used in cellular communication





Bit and Baud Rate Comparison

Modulation	Units	Bits/Baud	Baud rate	Bit Rate
ASK, FSK, 2-PSK	Bit	1	N	N
4-PSK, 4-QAM	Dibit	2	N	2N
8-PSK, 8-QAM	Tribit	3	N	3N
16-QAM	Quadbit	4	N	4N
32-QAM	Pentabit	5	N	5N
64-QAM	Hexabit	6	N	6N
128-QAM	Septabit	7	N	7N
256-QAM	Octabit	8	N	8N

QAM Example

- Example 1: A constellation diagram consists of eight equally spaced points on a circle. If the bit rate is 4800 bps, what is the baud rate?
  - The constellation indicates 8-PSK with the points 45 degrees apart.
  - Since $2^3 = 8$, 3 bits are transmitted with each signal unit.
  - Therefore, the baud rate is

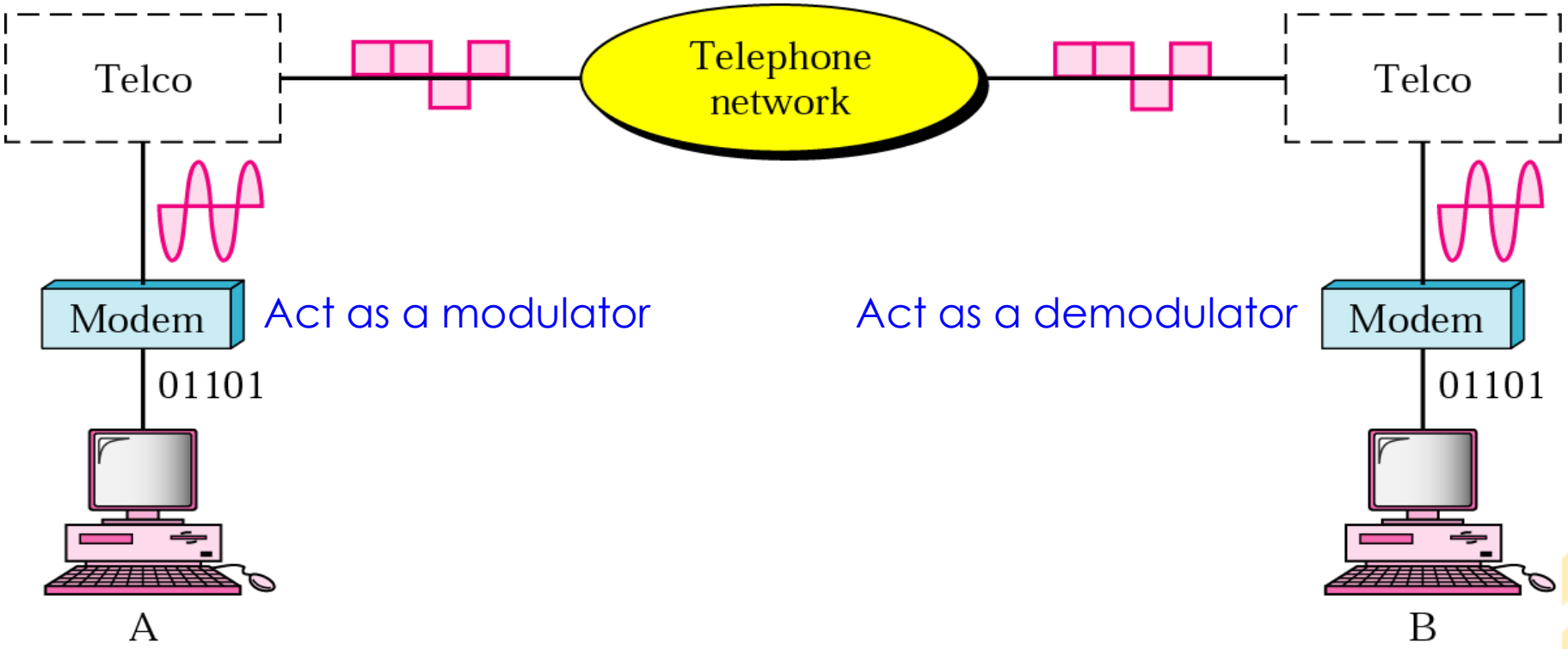
- Example 2: What is the bit rate for a 1000 baud 16-QAM signal?
  - The bit rate is

- Example 3: What is the baud rate for a 72,000 bps 64-QAM signal?
  - The baud rate is

Modem

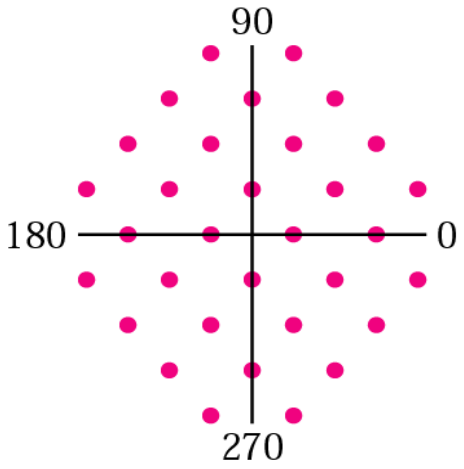
- Modem is a device which modulate and demodulate signals from computer in order to send the signal over the medium which is shared among users (e.g., air, coaxial or fiber optic cables).

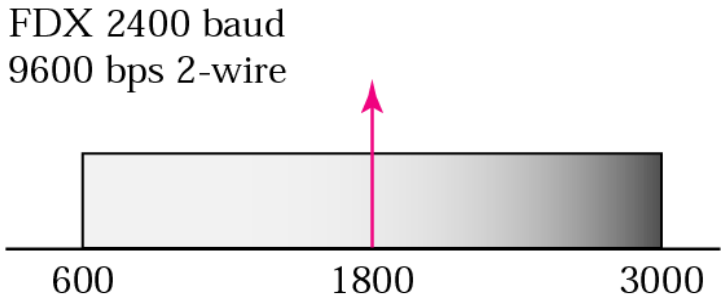
- Modem = modulator/demodulator

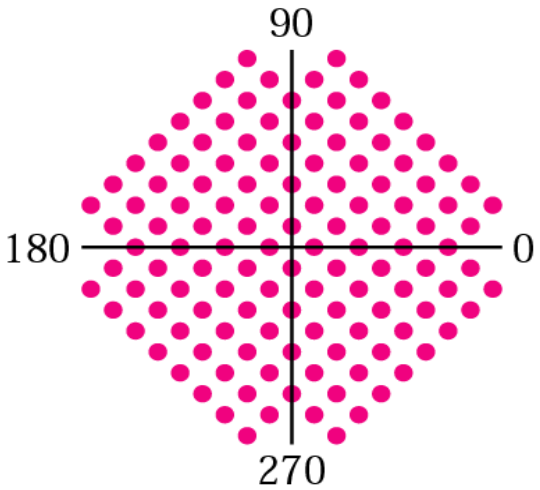


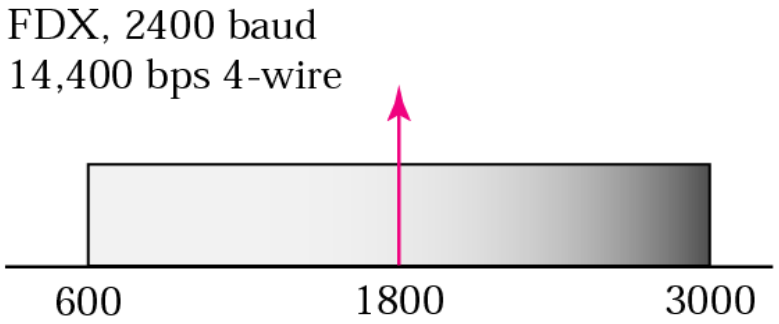
Telephone Modems (Obsolete)

- 20 years ago, computer signals are modulated into telephone wires.
- There were V.32 and V.90 modem standards using 32-QAM and 256-QAM.



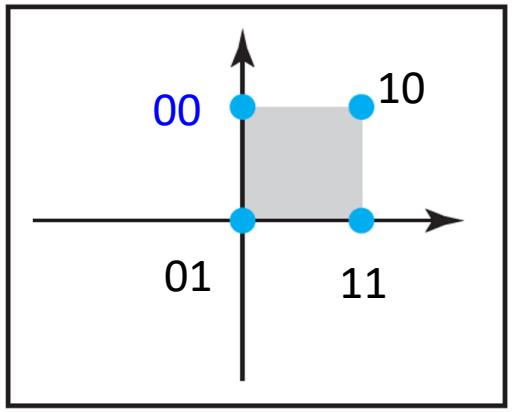




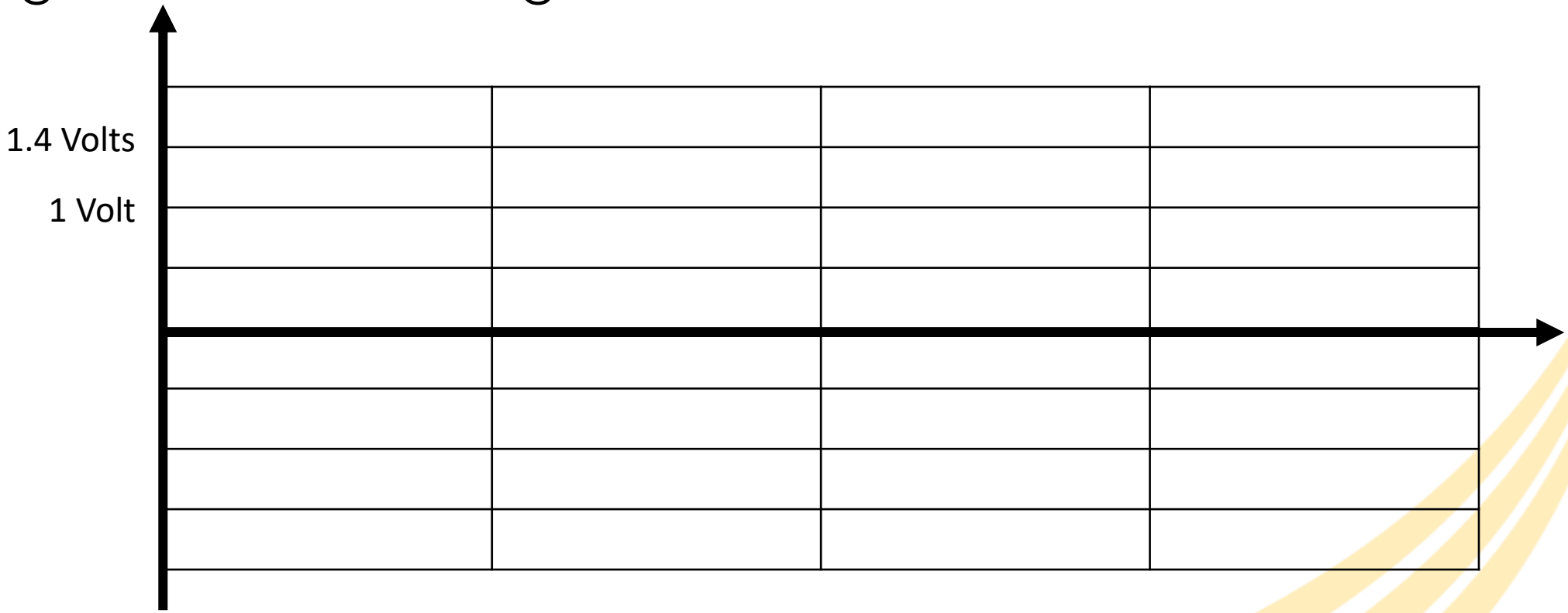


However, 256-QAM is still used in WiFi5
WiFi6 uses 1024-QAM (see Wiki for WiFi6)
Gigibit Wireless

QAM Exercise 1



- Assume the frequency is 1 Hz and Max amplitude is $\sqrt{2}$ volts (around 1.4), draw the signal graph when the sender what to send data "**00111001**" by using 4-QAM encoding as shown in the Figure.

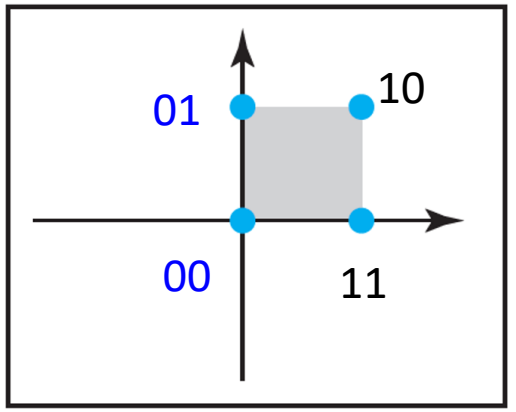


Assumption for QAM Exercise 1

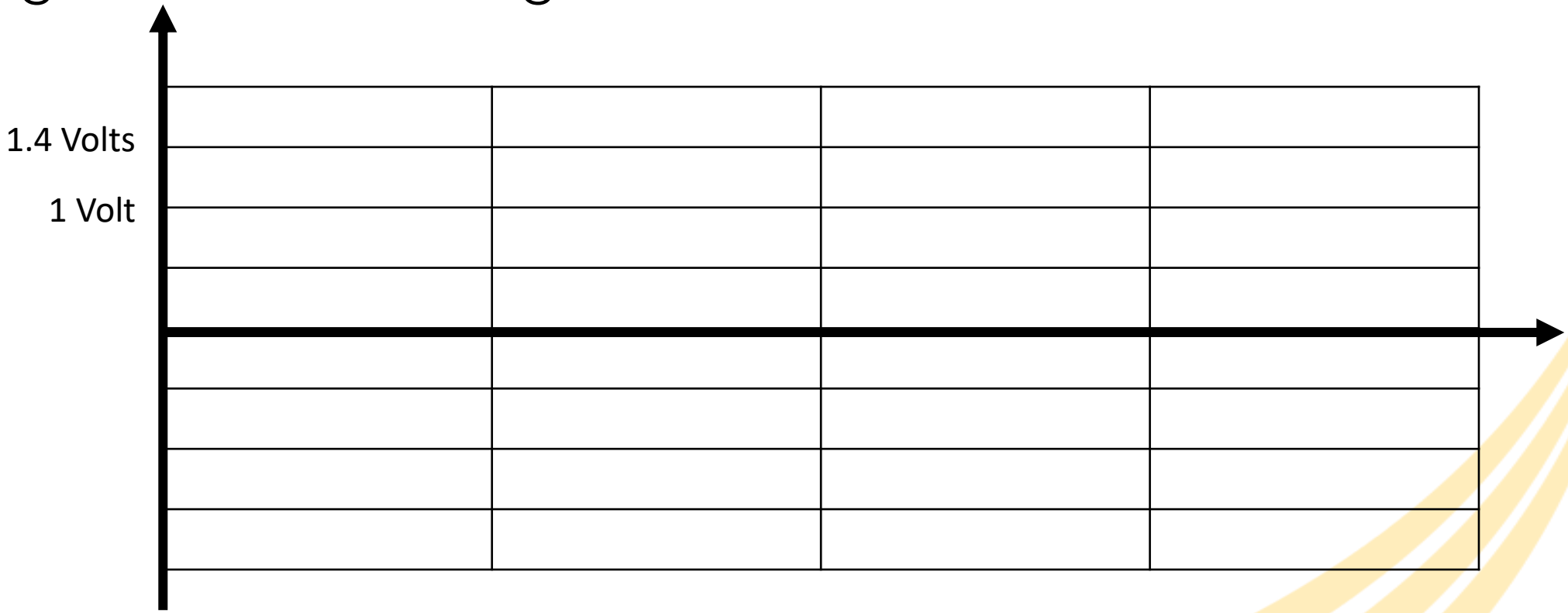
Assume one slot in your graph is 1 sec as well, so for 1 Hz, you can draw only one cycle in a slot.

But if I change the assumption to one slot = 2 sec, you have to draw 2 cycles in one slot.

QAM Exercise 2



- Assume the frequency is **2 Hz** and Max amplitude is $\sqrt{2}$ volts (around 1.4), draw the signal graph when the sender what to send data "**10110101**" by using 4-QAM encoding as shown in the Figure.



ITCS217 DataCom
Quiz #2 (Lectures 4 & 5 & 6)

Wed 26 & Thu 27 August 2026

Be honest and Do it by Yourself!

Sec1 & 3 : Thu 17 September 2026, 13:00-13:15

Sec2: Wed 16 September 2026, 9:00-9:15

Paper-Based
Prepare a pen

